Corporate Social Responsibility and Environmental Management

WILEY

Corporate Social Responsibility and Environmental Management

REVIEW ARTICLE OPEN ACCESS

The Circularity Paradox: The Rebound Effect, Governance, and the Limits of Corporate Sustainability

Taiz Viviane Dos Santos[1] | Sandra Marilce Diavon Alvez[1] | Simone Sehnem[2] | Dulcimar José Julkovski[1]

[1]Universidade do Oeste de Santa Catarina, Chapecó, Brazil | [2]Universidade do Oeste de Santa Catarina and Universidade do Sul de Santa Catarina, Chapecó, Brazil

Correspondence: Simone Sehnem (simonesehnem_adm@yahoo.com; simone.sehnem@unoesc.edu.br)

Received: 19 May 2025 | **Revised:** 29 August 2025 | **Accepted:** 10 September 2025

Funding: This study was supported by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), Fapesc – Fundaçãod e Ampara e Pesquisa do Estado de Santa Catarina, and the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES). The Article Processing Charge for the publication of this research was funded by CAPES (ROR ID: 00x0ma614).

Keywords: circular economy | consumption practices | critical assessment | multilevel governance | rebound effect | sustainability

ABSTRACT

The circular economy (CE), often celebrated as a sustainable panacea, faces increasing criticism due to the rebound effect (RE)—a paradoxical phenomenon in which efficiency gains indirectly lead to an increase in resource consumption. This study fills a relevant gap by analyzing, through a systematic review, how contemporary literature discusses the REs in circular strategies between 2015 and 2025. After screening the bibliographic portfolio, 75 relevant articles indexed in Scopus and Web of Science remained. The main results indicate that the RE is present in several circular strategies—such as reuse, recycling, remanufacturing, and rental models—and manifests itself through direct, indirect, psychological, and systemic mechanisms. Such effects are intensified by the absence of multilevel governance, appropriate indicators, and the diffusion of circular practices decoupled from cultural changes. The literature also points out that technological innovations, when devoid of regulation and sufficiency criteria, tend to legitimize overconsumption. The conclusion of the research reinforces that CE, if operated under a technocratic logic and based only on efficiency, runs the risk of reproducing the same unsustainable patterns of the linear model. Overcoming this paradox requires the articulation of robust governance, education for sufficiency, inclusive innovation, and specific metrics to monitor rebounds. As a groundbreaking contribution, this study proposes an original integrative framework—unparalleled in prior reviews—that dynamically interlinks circular strategies, RE types, conditioning factors, and mitigation instruments, drawing from an unprecedented multifactorial analysis of 75 articles to challenge the technocratic CE narrative and advance a regenerative, interdisciplinary agenda for true sustainability, with direct implications for policy, management, and future research.

1 | Introduction

The circular economy (CE) has established itself as a disruptive paradigm in the sustainability debate, challenging the linear logic of production and consumption through strategies such as recycling and reuse (Provensi and Sehnem 2025). Its main objective—to decouple economic growth from the use of natural resources (Geissdoerfer et al. 2017)—is based on efficiency, regeneration, and systemic innovation (Fraser et al. 2024; Skene 2018). However, critics question its effectiveness despite its biophysical and behavioral limits, putting pressure on its practical application (Corvellec et al. 2022).

One of the main challenges is the rebound effect (RE), a phenomenon in which efficiency gains encourage new consumption cycles, canceling out the environmental benefits. Studied in the field of energy efficiency (Saunders 1992; Sorrell 2009), it applies to CE when there are no robust cultural or political

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Corporate Social Responsibility and Environmental Management* published by ERP Environment and John Wiley & Sons Ltd.

changes. This dynamic can compromise the expected results of circularity, requiring more critical approaches. Recent literature warns of the risk of the RE, namely, efficiency gains in circular strategies (recycling, reuse, and remanufacturing) can, paradoxically, stimulate increased consumption and environmental and social externalities (Sorrell 2009; Murray et al. 2017). From the perspective of corporate social responsibility (CSR), the RE challenges the authenticity of business sustainability practices. Companies can adopt circular practices and still generate unintended negative impacts, such as increased total consumption, export of waste to developing countries, or precarious employment in the circular chain (Bocken et al. 2016; Geng et al. 2019). Thus, CE, if guided only by efficiency, can serve as greenwashing and compromise ESG commitments. Thus, the research question arises: how does the contemporary literature discuss RE in circular strategies between 2015 and 2025? This study proposes a critical analysis of the contemporary literature on RE in the CE, exploring systemic risks, theoretical controversies, and practical limitations. Based on a systematic literature review (SLR) based on Tranfield et al. (2003), 75 articles (2015–2025) from the Scopus and Web of Science databases were analyzed. Scopus and WoS are the largest databases and index the largest number of business and management journals. That is why they were chosen for this SLR. The research seeks to map theoretical and sectoral approaches, highlighting gaps and contradictions on the topic.

Although the CE has emerged as a transformative paradigm aimed at decoupling economic growth from resource depletion through strategies like reuse, recycling, and remanufacturing, it is increasingly critiqued for the RE—a paradoxical outcome where efficiency gains inadvertently spur heightened resource consumption and environmental externalities, undermining sustainability goals (Figge and Thorpe 2019; Bauwens 2021; Schultz 2022; Schultz et al. 2024). This challenge intersects profoundly with CSR, as businesses adopting circular practices may inadvertently perpetuate overconsumption, waste export to developing nations, or precarious labor conditions, thereby risking greenwashing and eroding the authenticity of their ESG commitments (Kjaer et al. 2019; Kirchherr 2022; Schultz and Pies 2024). An SLR in this domain is essential to synthesize fragmented insights from diverse fields, map theoretical controversies and empirical gaps—such as the under-explored interplay between multilevel governance, cultural shifts, and rebound mechanisms—and validate the research question's merit by providing a comprehensive, evidence-based foundation for rethinking CE beyond technocratic efficiency, toward truly regenerative models that align with CSR imperatives.

The approach becomes relevant considering the advancement of circular policies and practices that often ignore the potential side effects of their own innovations. As demonstrated by Saha et al. (2024), strategies such as product rental and reuse can, paradoxically, stimulate the intensification of consumption. Metic et al. (2024) highlighted that practices such as recycling can generate new forms of social exclusion or ecological overload. This paradox highlights the need for a critical review of the promises of circularity.

The following research questions are proposed for inclusion in the introduction: how does contemporary literature (2015–2025) discuss the RE in CE strategies, including its types, mechanisms, and implications for sustainability? What are the key conditioning factors—such as governance deficiencies, inadequate indicators, and cultural decoupling—that intensify REs in circular practices like reuse, recycling, and remanufacturing? In what ways does the RE challenge the authenticity of CSR practices, and what mitigation instruments, including robust governance and sufficiency education, are proposed to address this paradox? The period from 2015 to 2025 was chosen because it represents the period in which the RE began to be studied in the field of the CE.

The expected contribution of this paper is to offer a critical and systematic analysis of the phenomenon of the RE in the CE, a highly relevant and current topic for sustainability. Specifically, the study fills a relevant gap by reviewing the contemporary literature (2015–2025) that discusses the RE in circular strategies, based on 75 articles selected from the Scopus and Web of Science databases, providing an updated and comprehensive overview of the topic. It presents a detailed categorization of the types of REs (direct, indirect, psychological, and systemic) and the mechanisms that trigger them, contributing to the understanding of the complexity and multifactorial nature of the phenomenon. It highlights how the RE can compromise the environmental objectives of the CE, showing that efficiency gains can paradoxically lead to increases in resource consumption and waste generation, challenging the traditional logic that efficiency necessarily implies a reduction in environmental impact. It highlights the challenges of multilevel governance, the absence of adequate indicators, and the lack of cultural changes as factors that intensify REs, indicating ways to overcome these obstacles. It proposes an integrative framework that relates circular strategies, types of REs, conditioning factors, and mitigation instruments, offering a theoretical and practical tool for researchers, managers, and policymakers. It contributes to the academic debate by warning that the CE, if operated under a purely technocratic logic and based only on efficiency, can reproduce unsustainable patterns of the linear model, reinforcing the need for approaches that include education for sufficiency, including innovation and specific metrics to monitor and mitigate rebounds. Finally, the paper provides a solid basis for rethinking the CE in light of the risks of the RE, promoting a more critical, interdisciplinary agenda oriented toward real and fair sustainability, with implications for contemporary science, policy, and management.

The paper is organized as follows. After this introduction (Section 1), concepts, characteristics, consequences, and impacts on the topic—RE and CE—are presented (Section 2). Section 3 demonstrates the research methodology. Section 4 presents the presentation and analysis of the data, including the discussion of the results, while Section 5 presents the concluding remarks.

2 | RE and CE

In the CE, the RE takes on complex and multifactorial contours, given that this approach simultaneously seeks efficiency, regenerateness, and shared value throughout the life cycle of products (Ciechelska et al. 2023). The promise of closing the loop is accompanied by strategies such as reuse, remanufacturing,

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

composting, and recycling—all potentially vulnerable to the side effects of increased consumption. When a recycled product becomes cheaper, for example, this can lead to its overuse, generating externalities that cancel out the initial sustainability gains (Mallari et al. 2024).

The RE refers to the phenomenon in which gains in energy, material, or economic efficiency paradoxically lead to increased resource consumption or the intensification of environmental and social externalities (Scarpellini 2022). Originating in the classic Jevons (1865) paradox, who observed that the more efficient use of coal in the Industrial Revolution resulted in an overall increase in its consumption, the concept was further developed by Khazzoom and Brookes in the following decades and systematized by Sorrell (2009) as one of the great challenges to sustainability.

The main characteristics of the RE in the CE involve the following: (i) direct effects, such as increased consumption due to lower costs of circular products or services (Saha et al. 2024); (ii) indirect effects, such as redirecting saved income to other consumption (Metic et al. 2024); (iii) psychological effects, such as the moral license that consists of the perception that sustainable consumption exempts the consumer from environmental guilt (Chen 2021); and (iv) economy-wide effects, where sectoral efficiency increases aggregate demand and generates systemic impact (Saha et al. 2024).

The consequences of the RE in the CE are broad. In environmental terms, there is a risk of increased emissions, resource depletion (Prieto-Sandoval et al. 2019), and inefficient substitutions—such as recycled materials that do not fully replace virgin materials (Gheewala 2024). Socially, there are distributive impacts, such as the export of waste to developing countries or the precariousness of jobs in informal recycling sectors (Chen 2021). Economically, rebounds can generate market imbalances, overproduction, or speculative bubbles in sectors such as circular fashion or construction (Safarzynska et al. 2023). Contemporary literature on the subject has expanded, mainly based on systematic reviews, bibliometric studies, and quantitative life cycle analyses. Saha et al. (2024), for example, mapped the RE in the textile sector, identifying that the popularization of rental and second-hand models stimulated a new consumption pattern. Rabiu and Jaeger-Erben (2022) applied the Theory of Social Practices to demonstrate that circular consumption often legitimizes unsustainable patterns by reinforcing the narrative of conscious consumption. Seminal studies, such as those by Saunders (1992), who outline the logical structure of the RE, and Sorrell (2009), who systematize its main types and implications, continue to be fundamental references. However, new theoretical frameworks have been integrated into the critical analysis of the CE, such as ecological economics, which questions the viability of relative decoupling (Spiller et al. 2024), and degrowth thinking, which proposes a deliberate reduction in production and consumption as a response to planetary limits (Freeman 2018).

Contemporary criticisms of the CE, such as those by Skene (2018), reinforce the idea that the CE, if poorly implemented, can operate as a technocratic palliative that perpetuates consumerism in a new guise. Korhonen et al. (2018) warned of conceptual fragmentation, excessive focus on waste, and the absence of clear social goals. In other words, circularity without sufficiency and social justice is an imperfect cycle. Bibliometric reviews also indicate the growing interdisciplinarity of approaches to the RE, with articulation between environmental engineering, applied social sciences, supply chain management, and public policies (Shinde et al. 2022). In addition, Bartoszczuk (2023) highlights the gap in integration between top-down (regulatory) and bottom-up (behavioral) approaches, suggesting that mitigating the rebound requires multi-scale, participatory, and inter-sectoral innovation. The RE in the CE is not just an operational anomaly but a structural challenge that puts the very coherence of the circular model to the test. Ignoring it means legitimizing an illusory green growth. Recognizing it, on the other hand, allows us to rethink the role of circularity in the light of regenerative, fair, and truly sustainable economic models—an urgent agenda for science, politics, and contemporary management.

The RE in the CE represents a fundamental paradox, in which efficiency gains in circular strategies, such as cost reductions or product life extensions (PLEs), lead to an unexpected increase in resource consumption, partially or completely offsetting the intended environmental and social benefits (Zink and Geyer 2017; Schultz et al. 2024). Originating from the Jevons (1865) paradox, the RE in the CE manifests itself through economic, behavioral, and systemic mechanisms that amplify aggregate demand. Among the main mechanisms, the price mechanism stands out, in which cost reductions in circular products encourage greater consumption, as observed in PLE models in the textile sector, where discounts on recycled items can increase production and emissions; the income mechanism, in which generated savings are reallocated to other consumption, potentially generating a rebound of up to 52% in emissions; the output mechanism, in which productive efficiency scales total production, offsetting environmental gains; the motivational mechanism, including moral licensing, which makes sustainable actions justify excesses in other behaviors; and the time mechanism, where efficiency gains free up time for additional consumption activities. These mechanisms are interconnected and can be modeled using system dynamics (SD), revealing feedback loops that exacerbate RE, especially in the absence of sufficiency policies. Recent authors, such as Mallari et al. (2024), use SD to quantify these mechanisms in a sustainable fashion case, demonstrating that factors such as consumer sensitivity to discounts and environmental cognition intensify RE, with potential losses of 52.27% in emissions reductions. Similarly, Guzzo and Pigosso (2024) identify mechanisms such as reinvestment, factor substitution, and re-spending, applied to circular economy business models (CEBMs), such as washing machines as a service. They map 26 economic and behavioral mechanisms using a RE identification tool, highlighting how CEBM patterns stimulate consumption factors that trigger these effects.

Cycle strategy categories, in turn, aim to close resource loops through approaches such as narrowing (reducing material use), slowing (prolonging use), and closing (recycling and recovery), but they often generate RE by failing to consider behavioral and systemic dynamics. Key categories include:

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(i) reuse and repair, which extend the lifespan of products but can lead to careless use or increased turnover through moral license; (ii) remanufacturing and refurbishing, which restore products to "as new" quality; however, price reductions can increase aggregate demand; (iii) recycling, focused on closing loops but subject to imperfect substitution, where recycled materials do not fully displace virgin materials, resulting in continuous extraction; (iv) product-service systems (PSS), such as rental or subscription, which decouple ownership from use but encourage overconsumption through affordability; and (v) integrated circular business models (CEBM), encompassing ecodesign and reverse logistics, which optimize value chains but generate RE via economies of scale or income reallocation (Nußholz et al. 2023; Pádua Pieroni et al. 2018; Bauer et al. 2020). Ferrante et al. (2023) categorize strategies into dimensions such as business models, circular manufacturing ecosystems, and product lifecycle management (PLM), identifying gaps in industrial applications and performance measurement, with an emphasis on how mechanisms such as price and imperfect substitution affect PSS and reverse logistics. Schultz et al. (2024) reconceptualize RE from an ordonomic perspective, proposing mitigation via multilevel governance in strategies such as PSS, citing that traditional approaches underestimate the social and economic impacts of CEBM. Metic et al. (2024) explore RE in PSS via SD simulations, highlighting behavioral mechanisms in categories such as product rental. These strategies, while promising, require integration with sufficiency education and robust metrics to avoid rebounds, promoting a regenerative CE.

3 | Methodological Procedures

This research consists of an SLR, following the premises of Tranfield et al. (2003), covering the planning, implementation, reporting, and dissemination stages. In the planning, the scope was defined as articles published between 2015 and 2025, indexed in Scopus and Web of Science, in the area of Business Management and Accounting, limited to original and review articles in English, Portuguese, and Spanish. Table 1 presents the constructs, keywords, and search strings used. The study sought to strictly follow the criteria established to meet the proposed objective. To define the search strings, we identified previous studies on the topic and conducted SLRs. Based on these previous studies, tests were performed to identify the results generated. Considering that the aim of the research is to understand how contemporary literature discusses REs in circular strategies between 2015 and 2025, we chose to pursue a narrower scope of investigation, as also done by Metic and Pigosso (2022) and Ferrante et al. (2023).

The search strings were included in a combined form in a single search combining the term and correlates of CE with the term and correlates of RE. The review stage began with the construction of the bibliographic portfolio on March 20, 2025, when the search in the databases was performed, totaling 133 articles. The exclusion criteria were: areas other than Business Management and Accounting; papers published in languages other than English, Portuguese, and Spanish; papers published before 2015; duplicate papers; papers not available in full; and papers outside the scope of the research.

TABLE 1 | Constructs, keywords, and search strings defined.

Aspects	Description
Constructs	Rebound effect Circular economy
Keywords	Rebound effects Circular economy
Research Strings	("rebound effects" OR "rebound phenomenon" OR "jevons paradox" OR "backfire effects" OR "indirect rebound effects" OR "direct rebound effects" OR "energy rebound effects" OR "resource rebound effects" OR "environmental rebound effects" OR "sustainability rebound effects" OR "behavioral rebound effects" OR "economic rebound effects" OR "consumption rebound effects") AND ("circular economy" OR "circularity" OR "circular model" OR "circular systems" OR "circular design" OR "circular business models" OR "circular supply chains" OR "circular practices" OR "circular innovation" OR "closed-loop systems" OR "cradle-to-cradle" OR "resource efficiency" OR "waste to resource" OR "circular transition" OR "sustainable production and consumption")

The inclusion criteria used were: Business Management and Accounting; English, Portuguese, and Spanish; Article and Review; Year: 2015 to 2025; The word "rebound effect" must appear in the title, abstract, or keywords; the article must mention types of RE and the impacts generated; and the article must mention sustainable circular business models.

The results obtained in each database were exported to a BibTex file and inserted into the Parsifal software, where one duplicate article was excluded, followed by reading the title, abstract, and keywords of 132 articles. This first stage of portfolio classification followed the inclusion and exclusion criteria presented in Table 2 and resulted in the selection of 85 articles. Of the 85 articles selected, 10 articles were not downloaded, as it was not possible to access their full content. These articles were requested directly from the authors through ResearchGate and email, but without a timely response. Therefore, these 10 articles were excluded from the sample. For the 75 articles downloaded, AI DeepSeek, ChatGPT, and Qualitative Research Data Analysis were used to perform data extraction and content analysis using the PRISMA method. The entire methodological process adopted is presented in Figure 1.

Figure 1 shows the Prisma Protocol adopted to conduct this research.

In summary, the protocol adopted for data extraction and data analysis using AI followed the following steps:

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Criteria for the inclusion and exclusion of articles.

Inclusion criteria	Exclusion criteria
Business Management and Accounting	Different areas of Business Management and Accounting
English, Portuguese, and Spanish	Papers published in languages other than English, Portuguese, and Spanish
Article and Review	Conference paper, Book Chapter, Short Survey, and Retracted
Year: 2015–2025	Paper published before 2015
Include the word "rebound effect" in the title, abstract, or keywords	Duplicate papers
The article mentions types of rebound effect and the impacts generated	Papers not available in full version
The article mentions sustainable circular business models	Papers outside the scope of the research

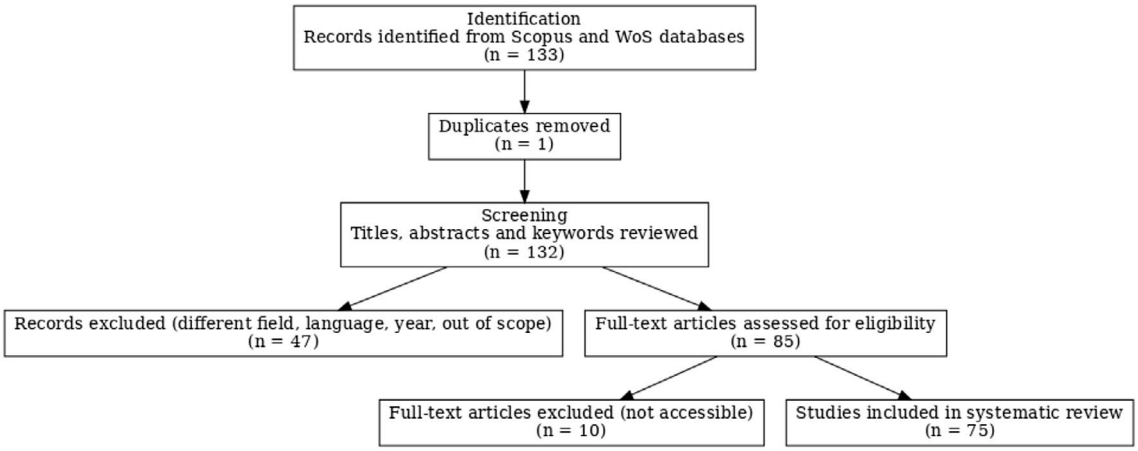

FIGURE 1 | PRISMA-based systematic review protocol.

- Defining the extraction criteria from the databases using specific search strings.
- Preparing the data for training: Downloading all papers in PDF format. Manually annotating relevant excerpts (labels) to serve as training data. Structuring the data in an AI-readable format (e.g., JSON and CSV);
- Preprocessing the text: Converting PDFs to text.
- Training the AI for data extraction: Using structured prompts to enable data extraction.
- Validation and refinement: Testing the model on a set of unseen articles; assessing accuracy by comparing it to human annotations; adjusting the prompts to the model as needed.
- Data collection and labeling: Data were extracted by attaching each of the analyzed papers one at a time.
- Effective prompting via human review and rechecking: Adjustments and refinements to the prompts were made via human review and rechecking.
- Manual validation to ensure reliability: Human review was performed to verify the reliability of the compiled data and its consistency with the papers analyzed. A similar protocol was already used by Sehnem et al. (2025) in a study that addresses the theme of Generation Z's Quiet Ambition.

Finally, it is worth highlighting that the objective of this study was to analyze, through a systematic review, how contemporary literature discusses REs in circular strategies between 2015 and 2025.

To facilitate the research, the Scopus and WoS databases were used. The data were exported to the Parsifal tool, which allows for the classification, screening, and removal of duplicate documents. They were subsequently exported in Excel format. The complete PDFs of the papers were downloaded from the databases. The AI tools DeepSeek and ChatGPT were used for data extraction. Multiple data extraction tools are recommended to ensure the reliability of the study. All of them served to attach the paper, input the prompt, and extract the data from each paper individually. The prompt used was the following: Prepare a detailed and complete summary in Portuguese of the attached paper. Next, extract the following information from the attached paper:

- Year the article was published.
- Journal where the article was published.
- Names of the paper's authors.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

- Country where the study was conducted.
- Objective of the study.
- Theory or theoretical approach used to develop the study.
- Concept of the RE is addressed in the study.
- Classical concept (Jevons and Khazzoom–Brookes) vs. contemporary approaches addressed in the study.
- Types of RE (direct, indirect, economy-wide, and psychological/behavioral) addressed in the study.
- Characteristics of the RE within the CE are addressed in the study.
- CE principles (closing loops, growth-resource decoupling, eco-design, or others) are addressed in the study.
- CE strategies (reuse, recycling, remanufacturing, lifespan extension, or others) addressed in the study.
- How CE initiatives can generate increased consumption/demand is addressed in the study.
- Sectors surveyed that have already mapped the RE in the implementation of the CE are addressed in the study.
- Priority industries that adopt the CE and that have identified a RE, as addressed in the study.
- Regulations that encourage the CE and their impact on aggregate consumption are addressed in the study.
- Public policies that encourage the CE and their impact on aggregate consumption are addressed in the study.
- How does the literature measure the RE in the CE, as addressed in the study?
- The role of innovation in advancing the CE and the resulting RE is addressed in the study.
- The role of Industry 4.0 technologies in advancing the CE and the resulting RE is addressed in the study.
- The role of governance in advancing the CE and the resulting RE is addressed in the study.
- The role of strategic stakeholders in advancing the CE and the resulting RE is addressed in the study.
- The role of networks, synergy, production chains, and the integration of actors and links in the chain for the advancement of the CE and the resulting RE is addressed in the study.
- Quantitative indicators of the RE in the CE are addressed in the study.
- Qualitative indicators of the RE in the CE are addressed in the study.
- Controversies and gaps in the RE in the CE are addressed in the study.
- Arguments in favor of the RE in the CE are addressed in the study.
- Critical arguments regarding the RE in the CE are addressed in the study.
- Instruments to reduce the RE in the CE are addressed in the study.
- Risks of the RE in the CE are addressed in the study.
- Types of REs associated with CE strategies are addressed in the study.
- Contradictory results mapped regarding the RE and the CE in the study.
- Does the analyzed study address the CE as a solution or a palliative for the RE? How is this done?
- Governance and actions needed to reduce the RE in the CE are addressed in the study.
- Agenda for future research on the RE in the CE is addressed in the study.
- Seminal articles on the RE in the CE are addressed in the study.
- Criticisms of the CE: Korhonen et al. (2018), Skene (2018), and others are addressed in the study. Detail what the study addresses in this aspect.
- Typology or taxonomy of the RE in the CE is addressed in the study.
- Limitations of the analyzed study.
- Recommendations for future work are addressed in the analyzed paper.

It is important to remember that this study presents partial data from the complete research that was conducted. The data extracted from each of the papers was pasted into a Word document. Once data extraction was complete, the Word document containing the extracted data was attached to Qualitative Research Data Analysis AI. At this point, the categorization and coding stage began using premises from Bardin (2015) for data categorization and coding. The findings were tabulated in summary tables that were duly analyzed using the content analysis proposed by Yin (2016). Table A1 presents the complete list of papers that were analyzed in this study.

4 | Data Presentation and Analysis

In this section, we present the data analysis, starting with the profile of the articles analyzed, and from this profile, we contextualize the main findings.

4.1 | Profile of the Articles Analyzed

Of the 75 articles, 57 articles are from the period 2021 to 2025. The journal with the largest publication was the Journal of Cleaner Production (10 articles), followed by the Sustainable Production and Consumption (8 articles), Sustainability (6 articles), and the Journal of Industrial Ecology (5 articles). The other publications are distributed in another 32 journals, among which the most prolific are: Journal of Environmental Management, Frontiers in Energy Research, Environmental

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Research Letters, and Frontiers in Sustainability, which were published in the period 2015–2025.

Different research sectors were identified, highlighting (1) textile/fashion, (2) energy, (3) transportation, (4) electronics/ICT, (5) food/agriculture, (6) sharing economy/retail, (7) construction, (8) bioeconomy/forestry, and (9) Copper Industry/Mining.

4.1.1 | Scope and Concepts Addressed in the Studies Analyzed

Table 3 presents a review of recent studies on the RE in the CE, highlighting the persistence of the classic concept of Jevons and Khazzoom–Brookes, which associates efficiency gains with increased consumption. Contemporary approaches broaden this perspective, exploring economic, behavioral, and systemic mechanisms such as prices, moral licensing, and SD.

There is a trend toward application in specific sectors, such as textiles, logistics, and agri-food, with diversified methods, including machine learning and econometric modeling. In addition, studies such as those by Chen (2021) and Shah and Zha (2025) incorporate social and macroeconomic dimensions, highlighting the complexity of the RE in the transition to sustainability. The table thus reveals a dialogue between classic theories and methodological innovations, pointing to the challenges in mitigating the undesirable impacts of CE.

Most studies (e.g., Saha et al. 2024; André and Björklund 2023; Mallari et al. 2024) reinforce that circular strategies, such as recycling, reuse, and sharing economy models, often lead to increased consumption due to factors such as price reductions, moral licensing, or expanded production scales. Jevons's classic concept (efficiency → increased consumption) is repeatedly mentioned, especially in sectors such as textiles, energy, and transportation.

Furthermore, it has been demonstrated that there is an expansion beyond the classic concept, with diverse methods (e.g., machine learning in Shinde et al. 2022; econometric modeling in Shah and Zha 2025) and integration of economic, behavioral, and systemic dimensions (e.g., moral licensing in Ciechelska et al. 2023; SD in Metic et al. 2024). This indicates a pattern of interdisciplinarity in the analysis of ER, focusing on sector-specific mechanisms.

The findings also reveal that ER varies by sector. For example, in the textile sector (Saha et al. 2024), ER is driven by lower prices for recycled products and consumption behaviors; in the energy sector (Shah and Zha 2025), macroeconomic effects predominate; while in the sharing economy (Laurenti et al. 2019), ER is linked to re-spending effects.

Therefore, Table 3 identifies the prevalence of the classic concept of the RE, which associates efficiency gains (in energy, materials, or processes) with a paradoxical increase in consumption, adapted to the context of the CE. For example, Saha et al. (2024) show that reduced prices for recycled products in the textile sector stimulate greater consumption, suggesting that companies should implement pricing strategies that internalize environmental costs (e.g., taxes on virgin resources, as suggested in Section 4.5). Methodological diversification (e.g., the use of SD in Metic et al. 2024) indicates that companies can adopt tools such as simulations and machine learning to predict and mitigate RE, dynamically adjusting inventories and prices to avoid overproduction. Bertassini et al. (2022) highlight the need for cultural alignment to avoid superficial practices that lead to greenwashing. Organizations should invest in internal education to promote a sufficiency mindset, reducing the impact of moral licensing. Circular fashion companies (e.g., clothing rentals) should monitor consumer behavior to avoid increased consumption due to moral licensing (Ciechelska et al. 2023). Strategies such as educational campaigns and design for durability can mitigate RE. Organizations in the energy sector can use macroeconomic modeling (Shah and Zha 2025) to anticipate increases in energy consumption due to efficiency gains, implementing regulatory policies (e.g., production limits). Platforms such as Airbnb or Uber must regulate access to avoid increasing aggregate consumption (Laurenti et al. 2019), using technologies such as blockchain for traceability (Rabiu and Jaeger-Erben 2022).

4.1.2 | Principles of the CE and Types of RE in the Studies Analyzed

Table 4 summarizes how different studies relate circularity strategies to possible REs, highlighting the diversity of approaches. The most frequent principles include closing loops, decoupling, and reuse, while the types of rebound range from direct and indirect impacts to behavioral and macroeconomic effects. Some studies, such as those by Chen (2021) and Ciechelska et al. (2023), extend the analysis to social dimensions, such as inequality and moral licensing. The table reveals that although the CE seeks sustainability, its systemic effects can generate unintended increases in consumption, requiring policies and models that mitigate these risks.

A highlight of Table 4 is the contrast between CE's positive intentions (sustainability through efficiency) and negative outcomes (unintentional increases in consumption). The papers analyzed describe these as systemic effects that can generate unintended increases in consumption, challenging the traditional logic that efficiency always reduces impacts. So, Table 4 demonstrates that the principles of the CE are intrinsically linked to different types of REs, requiring specific strategies to mitigate them. This relationship is explored in Table 5, which details the practices adopted by the authors, such as reuse, recycling, and product-service models (PSS), aiming to reduce unwanted impacts.

Evidence shows a link between CE principles (such as closing loops, decoupling, and reuse) and the occurrence of multiple types of ER (direct, indirect, psychological/behavioral, and economic-scale), which often compromise the expected environmental benefits of CE. Specifically, the studies reviewed (e.g., Saha et al. 2024; André and Björklund 2023; Mallari et al. 2024) highlight that core CE principles, such as closing loops (recycling, remanufacturing) and decoupling economic growth from resource use, are largely associated with increased consumption. For example, strategies such as recycling and reuse, although designed to reduce environmental impacts, often generate ER due to lower prices or perceived

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Scope and concepts addressed.

Authors	Scope	Classical concept (Jevons and Khazzoom–Brookes)	Contemporary approaches
Saha et al. (2024)	Critical review of the literature on CE in the textile sector, focusing on the rebound effect and governance.	Jevons and Khazzoom–Brookes: Efficiency → increased consumption.	Applied to CE: the impacts of strategies such as recycling and circular fashion.
André and Björklund (2023)	Analysis of the use phase and rebound effect in life cycle assessments in CE in the reuse of coats.	Jevons and Khazzoom–Brookes: Energy efficiency and paradoxes.	CE: Economic (price) and behavioral (novelty) mechanisms.
Shinde et al. (2022)	Assessment of the rebound effect on domestic consumption via machine learning applied to cooperative housing.	Mentioned: Efficiency → higher consumption (aggregate macroeconomic).	Bottom-up approach with machine learning; focus on households.
Metic et al. (2024)	Simulation of the rebound effect in circular fashion product-service systems (PSS) using system dynamics.	Jevons and Khazzoom–Brookes: Energy use.	CE: Systemic and behavioral responses (environmental motivation, price).
Mallari et al. (2024)	System dynamics model to analyze the rebound effect in circular strategies (reuse, recycling).	Jevons and Khazzoom–Brookes: Efficiency → consumption.	Specific effects of CE: Price, income, and economies of scale.
Freeman (2018)	Theory on the future of the rebound effect in scenarios of natural resource scarcity and socioeconomic change.	Jevons: Efficiency → increased usage.	Macro-effects: Broad economic, transformational, frontier, and so on.
Rabiu and Jaeger-Erben (2022)	Review of circular consumption practices and appropriation mechanisms, including the rebound effects.	Jevons: Efficiency results in higher consumption.	Sociostructural and behavioral factors (SPT).
Spiller et al. (2024)	Integration of circularity, efficiency, and sufficiency in nutrient management in agri-food systems.	Not addressed.	Reduction in systemic efficiency due to reuse of lower-quality nutrients.
Laurenti et al. (2019)	Mapping research on the sharing economy and its environmental and behavioral rebound effects.	It does not explore Jevons or Khazzoom–Brookes directly.	Behavioral re-spending effect in CE and the sharing economy.
Bartoszczuk (2023)	Critical analysis of CE limitations with a focus on RE and gaps in metrics and governance.	Jevons: Efficiency → higher consumption.	Social and technical, macroeconomic and sectoral dimensions of CE.
Ciechelska et al. (2023)	Empirical investigation of the rebound effect in the consumption of second-hand clothing, focusing on the price and moral effects.	Jevons and Khazzoom–Brookes: Rebound in energy.	Sectoral and psychological mechanisms: The price effect and moral licensing.

(Continues)

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | (Continued)

Authors	Scope	Classical concept (Jevons and Khazzoom–Brookes)	Contemporary approaches
Fertner and Große (2016)	Analysis of synergies and trade-offs in compact cities with a focus on resource efficiency and the rebound effect.	Jevons: Efficiency leads to higher consumption in other contexts.	Behavioral and indirect effects in sustainable urbanism.
Bertassini et al. (2022)	Assessment of the cultural readiness of companies to implement the CE, focusing on organizational maturity and the use of fuzzy logic.	There is no explicit citation of Jevons or Khazzoom–Brookes.	Emphasis on mitigating RE through cultural alignment and radical innovations.
Chen (2021)	Analysis of the environmental and social RE of the CE in sustainable cities, with a proposal for a classification framework.	The Jevons paradox: Efficiency improvements lead to higher consumption.	Expansion to include social effects (inequalities, health, and exclusion) in the CE.
Shah and Zha (2025)	Macroeconomic estimates of the energy rebound effect in the BRICS countries, considering the role of environmental policies and economic uncertainties.	Khazzoom–Brookes: Energy efficiency leads to increased energy demand.	Use of moderators such as environmental policies and economic uncertainties with econometric modeling.
Gheewala (2024)	Systemic vision of material recycling in the CE, considering the life cycle of the materials and the associated rebound effects.	Jevons: Efficiency gains result in increased consumption.	Emphasis on the systemic effects of the CE, such as imperfect substitution and incentives for consumption.
Safarzynska et al. (2023)	Simulation of recycling scenarios to assess how the CE can mitigate the material rebound effect during the energy transition.	Jevons: Efficiency leads to more consumption.	Material rebound in CE caused by renewables and limited recyclability; macroeconomic simulations.
Jaehn and Meissner (2022)	Investigation of the rebound effect in transport networks based on logistics decisions using the Traveling Purchaser Problem (TPP) model.	Jevons and Khazzoom–Brookes: Greater efficiency can generate more consumption.	Microeconomic focus on the rebound effect in logistics decisions (TPP) with backfire simulations.

sustainability, as observed in Saha et al. (2024) and Ciechelska et al. (2023). Table 4 shows that REs vary by sector. In the textile sector (Saha et al. 2024), ER is driven by prices and moral licensing; in the energy sector (Shah and Zha 2025), economic scale effects predominate; in the sharing economy (Laurenti et al. 2019), behavioral re-spending is central. This indicates that, although ER is a common phenomenon, its manifestation depends on the sectoral context.

The pattern of direct ER (e.g., increased consumption due to lower prices for recycled products, Saha et al. 2024) suggests that organizations should implement pricing strategies that prevent overproduction, such as taxes on virgin resources or dynamic pricing based on environmental impacts (Section 4.5). For example, textile companies can increase the prices of recycled clothing to reflect real costs, discouraging overconsumption. Principles such as reusability and lifespan extension (Ciechelska et al. 2023) require designs that prioritize durability and repairability, reducing psychological ER. Electronics companies, for example, can invest in modular products to minimize rapid replacement. Behavioral RE (e.g., Chen 2021) implies that companies should promote transparency about the real impacts of circular practices, avoiding sustainability narratives that legitimize excessive consumption. Educational campaigns can reinforce consumer awareness.

Companies should monitor consumption in second-hand markets, where psychological RE is prevalent (Ciechelska et al. 2023). Strategies such as rental caps or repair-based loyalty programs can reduce overconsumption. Economy-scale RE (Shah and Zha 2025) suggests that energy companies

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

should align efficiency gains with demand reduction policies, such as progressive tariffs or incentives for renewable sources. Platforms such as rental platforms (Metic et al. 2024) should regulate access to prevent increased use, integrating tracking technologies (e.g., blockchain; Rabiu and Jaeger-Erben 2022) to monitor impacts. Adopt metrics such as replacement rates (e.g., 1 new part for 1.23 used ones; Ciechelska et al. 2023) or cumulative environmental impact (Mallari et al. 2024) to assess RE in real time. As suggested by Bartoszczuk (2023), companies can form industrial symbiotic partnerships to transform waste into inputs, reducing resource extraction and indirect RE. Tools such as machine learning (Shinde et al. 2022) and SD (Metic et al. 2024) can predict and mitigate RE by adjusting production and consumption strategies.

4.1.3 | CE Strategies in the Studies Analyzed

It is observed that strategies such as lifespan extension (Ciechelska et al. 2023) and multiple-cycle recycling (Safarzynska et al. 2023) seek to balance efficiency and sustainable consumption. Together, the tables highlight the need to align principles, effects, and actions to ensure that the CE fulfills its potential without generating environmental setbacks.

Having presented the data analyzed in the literature, it is necessary to understand the mechanisms that lead to the RE. These mechanisms explain how initiatives such as reuse and recycling can, paradoxically, increase consumption due to economic, behavioral, and systemic factors. Analyzing these mechanisms is essential to develop policies and business models that minimize undesirable effects, ensuring that the environmental benefits of the CE are fully achieved.

4.2 | RE Mechanisms in CE

Studies have shown that RE mechanisms can lead to increased consumption/demand due to cost reduction, moral license, and efficiency in production processes. Regarding cost reduction, studies show that circular practices, such as recycling, for example, end up encouraging more consumption because they reduce costs. Saha et al. (2024) warned that cheaper recycled materials in the textile sector lead to higher production. Mallari et al. (2024) confirmed that price reductions generate a RE via increased direct and indirect consumption. Regarding moral license, Ciechelska et al. (2023) point out that those who buy used clothes tend to buy more, believing they are doing good. Metic et al. (2024) indicated that renting clothes leads to increased use for the same reason. André and Björklund (2023) associate this behavior with the desire for novelty. In this sense, we found that consumers feel more comfortable and authorized to consume more because they believe they are consuming sustainably.

As for process efficiency, we can say that even with less impact per unit, the total production can increase; that is, less energy per unit is compensated by increased production. In this sense, Saha et al. (2024) showed that efficiency gains can expand the scale of production. Along the same lines, Spiller et al. (2024) showed that nutrient reuse, if less efficient, can increase environmental losses.

In Table 6, we present the RE mechanisms in the CE, addressed in the studies analyzed.

Table 6 reveals that the mechanisms of RE in the CE operate across multiple dimensions, from direct economic factors (such as price reductions and income reallocation) to behavioral and systemic aspects. Psychological effects, such as moral licensing (Ciechelska et al. 2023) and the false sense of sustainability (Bertassini et al. 2022), which encourage overconsumption, stand out.

In addition, macroeconomic mechanisms (Freeman 2018) and inappropriate substitutions (Gheewala 2024) demonstrate how efficiency gains can generate aggregate increases in resource use. These findings reinforce the need for policies that consider these complexities to prevent the CE from resulting in undesirable environmental impacts. The next section examines the main methodologies and quantitative and qualitative approaches used by researchers to measure and evaluate the RE phenomenon in CE initiatives.

4.3 | How Does the Literature Measure the RE in CE?

The literature measures the RE in the CE through quantitative (Table 7) and qualitative (Table 8) indicators, demonstrating methodological diversification to measure the RE in the CE.

Table 7 presents the quantitative indicators used to measure the RE in the CE, highlighting metrics such as GWP reduction (4% with recycled cotton) and replacement rates (1 new item for every 1.23 used items), as well as models such as LCA and systemic simulations. These data reveal a technical approach focused on material and energy efficiency, but also expose gaps in capturing behavioral and social aspects. This limitation is addressed by Table 8—Qualitative Indicators, which explores perceptions, motivations, and cultural practices, such as moral licensing and social practices theory (SPT), essential to understanding the indirect mechanisms of the RE. Together, the tables demonstrate the need to integrate quantitative and qualitative analyses for a holistic assessment of the impacts of the CE.

The results presented in Tables 7 and 8 reveal both methodological advances and persistent limitations in measuring the RE in the CE. The next section critically examines the points of controversy and gaps in the RE in the CE. Table 9 shows the indicators mapped.

4.4 | Controversies and Gaps in RE in CE

In Table 10, we present some controversies in the RE in the CE and their impacts.

The controversies presented in Table 9 indicate that the RE is a risk to the effectiveness of the CE, examples such as: reuse leading to an increase in total consumption (André and Björklund 2023; Rabiu and Jaeger-Erben 2022); rental and sharing platforms increasing emissions and consumption (Metic et al. 2024; Laurenti et al. 2019); recycling and efficiency

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | Principles of the CE and types of RE.

Authors	CE principles addressed	Types of rebound effect covered
André and Björklund (2023)	Closing cycles, Decoupling.	Direct, Indirect, Behavioral.
André and Björklund (2023)	Slowing: Reuse/repair/life extension	Indirect
Arpin et al. (2024)	Digitization	Indirect
Bączyk et al. (2024)	Digitization	Indirect
Bartoszczuk (2023)	Closing cycles, 3Rs, Ecodesign.	Direct, Indirect, Economy-wide.
Bertassini et al. (2022)	Closing cycles (recycling), decoupling growth-resources.	Psychological/behavioral—Cultural Misalignment creates a false sense of sustainability and increased consumption.
Beyeler and Jaeger-Erben (2022)	Digitization, Sufficiency	Indirect, Mitigation via sufficiency
Birat (2015)	Closing: Closing/recycling/remanufacturing; Narrowing: resource/energy efficiency	No mentioned
Bocken et al. (2022)	Digitization, Sufficiency	Indirect, Mitigation via sufficiency
Bubinek et al. (2025)	Digitization; Slowing: Reuse/repair/life extension	Indirect
Castro et al. (2022)	Narrowing: Resource/energy efficiency	RE (unspecified)
Chatzistamoulou and Tyllianakis (2022)	Digitalization	Indirect
Chatzistamoulou and Koundouri (2024)	Narrowing: Resource/energy efficiency	RE (unspecified)
Chen (2021)	Closing cycles (recycling, remanufacturing), decoupling growth-resources.	Direct, indirect, economy-wide, and psychological/behavioral—also focusing on social inequality and occupational health.
Ciechelska et al. (2023)	Closing cycles, Reuse, Extending useful life.	Direct (price effect), Psychological/Behavioral (moral licensing).
Csutora et al. (2022)	Digitalization	Indirect
Das et al. (2023)	Closing: Closing/recycling/remanufacturing; Digitalization/ICT	Indirect
Fan et al. (2023)	Digitalization	Indirect
Ferrante et al. (2024)	Narrowing: Resource/energy efficiency	RE (unspecified)
Fertner and Große (2016)	Decoupling and reuse of urban land (brownfield development).	Direct, Indirect, Behavioral.
Figge and Thorpe (2019)	No mentioned	RE (unspecified)
Flachenecker (2018)	Governance/Policies/Frameworks	Macroeconomic
Flores et al. (2022)	Digitalization	Digital
Font Vivanco et al. (2018)	Digitalization	Digital
Font Vivanco et al. (2022)	Digitalization	Digital
Freeman (2018)	Closing cycles, Industrial symbiosis.	Broad, Transformational, Frontier, International Economics.
Genovese et al. (2024)	No mentioned	RE (unspecified)
Gheewala (2024)	Closing cycles (recycling, reuse), decoupling via efficiency and dematerialization.	Direct, indirect, and economy-wide—increased demand and imperfect substitution of recycled materials.

(Continues)

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | (Continued)

Authors	CE principles addressed	Types of rebound effect covered
Godinot et al. (2024)	Digitalization	Digital
Gossart (2014)	Digitalization	Digital
Grădinaru and Maricuț (2022)	No mentioned	RE (unspecified)
Henry et al. (2021)	PSS/Sharing	Sharing (convenience/time)
Huang and Lin (2025)	Digitalization	Digital
Jaehn and Meissner (2022)	Do not address CE.	Direct logistical decisions generate increased consumption despite greater efficiency in transportation.
Kjaer et al. (2018)	Governance/Policies/Frameworks; Narrowing: Resource/Energy Efficiency	No mentioned
Koide et al. (2022)	Digitization/ICT; PSS/Sharing	Digital/ICT; PSS (intensive use)
Konash and Nasr (2022)	Narrowing: Resource/energy efficiency	No mentioned
Lahcen et al. (2024)	No mentioned	RE (unspecified)
Laurenti et al. (2019)	Property access, Reuse.	Behavioral/Psychological (re-spending rebound effect).
Levänen et al. (2021)	Closing: Closing/recycling/ remanufacturing	No mentioned
Liu et al. (2022)	Closing: Closing/recycling/ remanufacturing; Digitalization/ICT	Digital
Lowe et al. (2024)	Narrowing: Resource/energy efficiency	RE (unspecified)
Makov and Font Vivanco (2018)	Slowing: Reuse/repair/life extension	Indirect
Mallari et al. (2024)	Closing cycles, Decoupling, Reuse, Recycling, Remanufacturing, Refurbishing.	Price Effect, Income Effect, Economies of Scale.
Mata et al. (2022)	Digitalization	Digital
Metic and Pigosso (2022)	Narrowing: Resource/energy efficiency	RE (unspecified)
Metic et al. (2024)	Closing cycles, Decoupling, Reuse, and PSS (Product-Service).	Direct, Indirect, Behavioral.
Morimoto et al. (2021)	Closing: Closing/recycling/ remanufacturing	RE (unspecified)
Niehoff and Beier (2018)	Digitalization	Digital
Niero et al. (2021)	No mentioned	RE (unspecified)
Onat et al. (2023)	No mentioned	RE (unspecified)
Ottelin et al. (2020)	No mentioned	RE (unspecified)
Piscicelli (2023)	Digitalization	Digital
Pohl et al. (2019)	Digitization/ICT; Narrowing: resource/energy efficiency	Digital
Rabiu and Jaeger-Erben (2022)	Closing cycles, extent types useful life.	Direct, Indirect, Psychological/ Behavioral.
Roa and Rosendahl (2023)	Digitalization/ICT; Governance/ Policies/Frameworks	Digital

(Continues)

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | (Continued)

Authors	CE principles addressed	Types of rebound effect covered
Safarzynska et al. (2023)	Closing cycles via recycling (single, multiple, and infinite).	Direct and economy-wide—cheaper recycled products and capital intensity increase aggregate material use.
Saha et al. (2024)	Closing cycles, Decoupling, Ecodesign.	Direct, Indirect, Economy-wide, Psychological/Behavioral.
Salvador et al. (2021)	Ecodesign/Durability; Narrowing: Resource/Energy Efficiency; Regeneration/Bioeconomy	No mentioned
Salvador et al. (2022)	Digitalization/ICT; Regeneration/Bioeconomy	Digital/ICT
Samani (2025)	No mentioned	RE (unspecified)
Santarius et al. (2018)	Governance/Policies/Frameworks	No mentioned
Schultz et al. (2024)	No mentioned	RE (unspecified)
Shah and Zha (2025)	Do not address the CE principles.	Economy-wide—aggregate increase in energy consumption, moderated by environmental policies and economic uncertainties.
Sheng et al. (2023)	Digitalização/ICT	Digital/ICT
Shinde et al. (2022)	Not directly applicable to CE, but related to resource efficiency.	Direct, Indirect.
Siderius and Poldner (2021)	Narrowing: Resource/energy efficiency	RE (unspecified)
Sourabh et al. (2024)	No mentioned	RE (unspecified)
Spiller et al. (2024)	Closing of nutrient cycles	Direct (e.g., substitution with less efficient products)
Tsuji et al. (2020)	Digitization/ICT; PSS/Sharing	Digital/ICT; Sharing (convenience/time)
Vegter et al. (2023)	Digitalization/ICT	Digital/ICT
Wallenborn (2018)	No mentioned	RE (unspecified)
Warmington-Lundström and Laurenti (2020)	Closing: Closing/recycling/remanufacturing; Digitization/ICT; PSS/Sharing	Digital/ICT; Sharing (convenience/time)
Wiprächtiger et al. (2022)	No mentioned	RE (unspecified)
Zerbino (2022)	Governance/Policies/Frameworks; Narrowing: Resource/Energy Efficiency	RE (unspecified)

not always generating net benefits (Spiller et al. 2024), reveal that the results of the CE are sometimes contradictory. The literature criticizes technocratic approaches that overestimate the benefits of CE without considering the real dynamics of use and consumption, aligning with the ecological criticism that relative efficiency does not guarantee absolute decoupling (André and Björklund 2023; Mallari et al. 2024; Metic et al. 2024).

Another aspect identified in previous literature is increased consumption as a dominant impact (Saha et al. 2024; André and Björklund 2023; Ciechelska et al. 2023). Impacts include stimulating consumption through low prices, accessibility, or legitimization (e.g., reuse stimulates new consumption; sharing reduces cost but increases usage). These impacts reinforce RE as a vicious cycle, where circular strategies (e.g., reuse, sharing) paradoxically intensify linear consumption patterns.

Above all, the focus on specific CE strategies (Reuse, Recycling, and Sharing) still prevails, as discussed by Rabiu and Jaeger-Erben (2022), Bartoszczuk (2023), and Laurenti et al. (2019), and generates behavioral or systemic impacts. Furthermore, flaws in the conceptual understanding of the CE often relegate RE to efficiency, and it should also prioritize sufficiency. Thus, we can conclude that a technocratic, palliative understanding of the CE prevails, resulting in increased consumption and environmental losses. This presents an opportunity to strengthen analytical thinking that integrates efficiency, sufficiency, and governance.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 | CE strategies.

Authors	CE strategies addressed
Saha et al. (2024)	Reuse, Recycling (textile fibers), Life extension (repairs), AB-PSS (rental models)
André and Björklund (2023)	Reuse (second-hand sale), Extending service life (repairs)
Metic et al. (2024)	Reuse (dress rental), Product-Service (PSS)
Mallari et al. (2024)	Recycling, Reuse, Remanufacturing, Refurbishing
Freeman (2018)	Reuse, Recycling, Waste reduction, Industrial symbiosis
Rabiu and Jaeger-Erben (2022)	Reuse, Repairs, Recycling, Remanufacturing
Spiller et al. (2024)	Nutrient reuse (e.g., organic waste recovery)
Laurenti et al. (2019)	Reuse (second-hand), Rental, Access without transfer of ownership
Bartoszczuk (2023)	Reuse, Recycling, Remanufacturing, Life Extension, Ecodesign
Ciechelska et al. (2023)	Reuse (second-hand clothing), Extending the useful life
Fertner and Große (2016)	Reuse of urban land (brownfield development)
Bertassini et al. (2022)	Recycling, reuse, remanufacturing, design for extended cycles
Chen (2021)	Recycling, remanufacturing, reverse logistics, design for extended cycles
Gheewala (2024)	Recycling (metals, plastics), reuse (e.g., glass bottles), and extending the useful life of products
Safarzynska et al. (2023)	Recycling with different cycles (single, double, infinite); focus on mitigating material rebound

The gaps presented in Table 11 indicate flawed planning, that is, the RE is not considered, and environmental gains from circular strategies are overestimated. These gaps impact society by deepening inequalities when implementing a CE that exports impacts on vulnerable populations or when more sustainable products encourage greater turnover. Thus, having identified the controversies and gaps, we will analyze the instruments to reduce the RE in the CE.

4.5 | Instruments to Reduce the RE

The studies analyzed present different instruments to mitigate the RE in the CE, highlighting approaches ranging from behavioral to technological (Tutore et al. 2024) and regulatory. Public policies such as environmental taxes, strict regulation, extended producer responsibility (Saha et al. 2024), fiscal incentives for reuse (André and Björklund 2023), virgin resource taxes, consumer education, standards such as BS 8001:2017 (Bartoszczuk 2023), limits on manure application and production quotas to avoid intensification in the agricultural sector (Godinot et al. 2024), and other subsidies for circular practices are recommended and aim to internalize environmental costs, reducing incentives for overconsumption.

Consultation and awareness (Saha et al. 2024; Ciechelska et al. 2023) are presented as key instruments in combating psychological effects, such as moral license (Lowe et al. 2024), to consume more after any sustainability action (Zerbino 2022).

Transparency in communicating the real impacts of CE initiatives, including rebound risks, to avoid greenwashing (Arpin et al. 2024; Zerbino 2022; Chen 2021).

Design innovation (Saha et al. 2024) and technologies such as blockchain for traceability (Rabiu and Jaeger-Erben 2022) are highlighted as facilitators, but require monitoring to avoid externalities such as job displacement. In addition, the integration of bottom-up and top-down models (Shinde et al. 2022) and multilevel and collaborative intra/inter-sectoral governance (Zerbino 2022) is considered essential to coordinate actions among companies, governments, and civil society.

However, the effectiveness of the rebound mitigation instruments varies according to the context. For example, restrictive policies have proven effective in the energy and agriculture sectors (Godinot et al. 2024; Chatzistamoulou and Koundouri 2024), while education-based initiatives require long-term cultural changes (Bączyk et al. 2024). In summary, mitigating the RE requires strategies applied to sectoral and behavioral dynamics that combine multiple factors, in addition to being aligned with economic, regulatory, and consumer awareness aspects.

4.6 | Theories That Can Be Integrated Into Studies to Mitigate the RE

Neo-institutionalism, corporate moral responsibility (CMR) theory, UTAUT, and dynamic capabilities theory were used by Saha et al. (2024) to analyze the theoretical assumptions of CE and RE in the textile sector. They highlight the risk of illusory decoupling and recognize systemic RE—in line with the ecological economy's critique of relative decoupling, as well as questioning the narrative that CE always generates benefits, which can be deepened by the degrowth critique of green capitalism.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 6 | RE mechanisms.

Authors	Rebound effect mechanisms in CE addressed
Saha et al. (2024)	Reduced price → increased consumption; perception of sustainability → "license to consume" (behavioral); structural increase in demand (economy-wide).
André and Björklund (2023)	User behavior (desire for novelty); reuse generates savings that are spent on new products.
Shinde et al. (2022)	Reduction of expenses (e.g., rent) → redirection of consumption to other sectors.
Metic et al. (2024)	Low price and environmental motivation → increased rental of products (dresses); moral license.
Mallari et al. (2024)	Price effect (R1): discounts increase consumption; income effect (R2): saved income redirected; economies of scale (B4) encourage Supply.
Freeman (2018)	Broad economic effects (general growth), transformational (creation of new industries), frontier, International.
Rabiu and Jaeger-Erben (2022)	Perceived efficiency leads to the reallocation of resources (indirect effect); circular practices legitimize additional consumption (psychological).
Spiller et al. (2024)	Reuse of lower efficiency nutrients → increased environmental losses.
Laurenti et al. (2019)	The re-spending effect: Income saved with EX is directed to other consumption.
Bartoszczuk (2023)	Greater efficiency → increased use of resources; circular practices that do not completely replace new products.
Ciechelska et al. (2023)	Price effect: Low cost of SHC → increased consumption; moral licensing → justification for consuming more.
Fertner and Große (2016)	Urban efficiency (e.g., transportation) leads to new consumption (e.g., recreational travel); urban trade-offs.
Bertassini et al. (2022)	Insufficient cultural alignment creates a false sense of sustainability, encouraging superficial practices (e.g., excessive focus on recycling).
Chen (2021)	Cost reduction and a sense of sustainability lead to overconsumption; waste export causes inequality and health risks.
Gheewala (2024)	Cheaper recycled materials increase demand; imperfect substitutions between recycled and virgin materials maintain the use of primary resources.
Safarzynska et al. (2023)	Cheap recycled products stimulate demand; finite recycling creates scrap scarcity, encouraging the extraction of new materials.

André and Björklund (2023) used a product-user framework and the RE theory. They show that user behavior can cancel out environmental gains, opening space to discuss the biophysical limits and the need for absolute decoupling. They also noted that reuse can increase total consumption. This can be criticized from a degrowth perspective as a failure of a market-based solution.

Metic et al. (2024) used SD, RECF (Conceptual Framework of the Rebound Effect) to simulate rebound dynamics in PSS systems (e.g., dress rental). The SD model reveals how local gains (dress rental) lead to systemic increases. This allows simulating the scenarios of absolute vs. relative decoupling. It also offers the opportunity to explore non-growth policies, such as limits on consumption or income, consistent with degrowth thinking.

Spiller et al. (2024) aimed to propose combined strategies to avoid rebounds in nutrient management and used the Material Flow Analysis (MFA) approach, integrating circularity, efficiency, and sufficiency. They proposed the integration of circularity, efficiency, and sufficiency—an ideal platform to introduce the concept of absolute decoupling. Furthermore, they support the idea that efficiency and circularity alone are not enough—in line with degrowth, which proposes reducing production and consumption as an explicit goal.

SPT and life cycle assessment (LCA) were used to investigate the appropriateness of circular practices and associated rebounds (Rabiu and Jaeger-Erben 2022). They highlight that circular practices often increase aggregate consumption—a criticism compatible with the defense of resource use limits, the focus of ecological economics. They advocated profound changes in social practices, which allowed the integration of the principles of sufficiency and planned reduction of consumption, typical of degrowth.

Mallari et al. (2024) used SD to model rebound based on price, income, and economies of scale effects. Modeling scale and price

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 7 | Quantitative indicators identified.

Authors	Mapped indicators
Saha et al. (2024)	They used life cycle analysis (LCA) to measure GWP (global warming potential) reduction, such as 4% with recycled cotton.
Shinde et al. (2022)	They applied random forest regression to estimate CO_2-eq per CHF spent, and the rebound factor was calculated by the difference between the expected and actual consumption.
Metic et al. (2024)	They simulate the magnitude of the rebound (10%–53%) and analyze the average use of products per consumer.
Mallari et al. (2024)	They measured the accumulated environmental impact (in kg-CO_2) and the difference between the potential and realized environmental benefits.
Ciechelska et al. (2023)	They calculated the replacement rate (1 new part replaced for 1.23 used ones) through statistical analysis with 639 students.
Spiller et al. (2024)	They assessed the nutrient use efficiency and reuse levels in the agricultural systems.
Bartoszczuk (2023)	Uses recycling and reuse rates, as well as LCA metrics.
Rabiu and Jaeger-Erben (2022)	They applied LCA to calculate the avoided emissions and life cycle efficiency.
Chen (2021)	Measures the percentage of renewable energy and indicators related to the ISO 37120 standard.
Jaehn and Meissner (2022)	Proposes a direct rebound formula: RE = 1 − (actual reduction ÷ expected reduction), applied to operational simulations.

TABLE 8 | Qualitative indicators.

Authors	Mapped indicators
André and Björklund (2023)	They investigated the purchase/disposal motivations, frequency of use, and behavioral perceptions of consumers and store managers.
Rabiu and Jaeger-Erben (2022)	They apply social practices theory (SPT) to capture the cultural meanings, routines, and skills associated with circular consumption.
Saha et al. (2024)	They used interviews to assess perceptions of hygiene in circular clothing and the understanding of the real impacts of CE.
Metic et al. (2024)	They analyzed the feedback loops and environmental motivation in the simulated scenarios.
Ciechelska et al. (2023)	They assess moral licensing and sustainable motivations that, paradoxically, encourage excessive consumption.
Laurenti et al. (2019)	They highlight the scarcity of LCA and resort to qualitative analysis of user motivations.
Freeman (2018)	It uses socioeconomic scenarios (SSPs) with a conceptual approach, without direct quantification.
Bertassini et al. (2022)	They used a questionnaire with fuzzy logic and a radar chart to identify behaviors that can generate a rebound.
Chen (2021)	Considers stakeholder participation, social impacts, and community perception in sustainable decisions.

effects aligns with ecological criticisms of the illusion of sustainability through efficiency. Ideal for applying degrowth scenarios, with limits to consumption growth even with efficiency.

To quantify RE in cooperative households, Shinde et al. (2022) used random forest regression (machine learning), a bottom-up approach. They emphasize that household efficiency policies can generate more consumption in other sectors, illustrating the recoupling effects, the focus of ecological economics. Although technocratic, the model can be confronted with planetary limits and demands for reduced consumption, consistent with

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 9 | Indicators mapped.

Authors	Quantitative indicators	Qualitative indicators	Distribution trend	Study type
Saha et al. (2024)	GWP reduction by 4% (recycled cotton)	Perception of hygiene in second-hand clothes	Direct, indirect, economy-wide, and behavioral effects in circular fashion	Empirical with indicators
André and Björklund (2023)	Frequency of use and lifespan of coats	Discarding motivations (desire for novelty)	Reuse can intensify consumption; strong behavioral effect	Empirical with indicators
Shinde et al. (2022)	CO_2-eq emissions per CHF spent; Rebound factor	Not applied	Direct and indirect effects on housing and domestic consumption	Empirical with indicators
Metic et al. (2024)	RE Magnitude: 10%–53%; average dress usage vs. Inventory	Feedback loop analysis (CLDs)	Higher RE in low price scenarios; wide range (10%–53%)	Empirical with indicators
Mallari et al. (2024)	Cumulative environmental impact (kg-CO_2); potential vs. realized benefits	Not mentioned	Price, income, and scale effects in secondary markets	Empirical with indicators
Spiller et al. (2024)	Replacement ratio 1:1.23 (SHC clothing)	Consumer motivations (economic, environmental)	Net increase in consumption due to imperfect substitution	Empirical with indicators
Bartoszczuk (2023)	Recycling rate (e.g., 38.7% waste Poland)	Preservation of product functions/value	Direct, indirect, and economy-wide effects on sectors such as packaging and electronics	Empirical with indicators

TABLE 10 | Controversies in RE in CE.

Controversies	Impacts	Authors
Overestimation of the benefits of EC.	Increased demand and cancellation of environmental gains.	Saha et al. (2024)
Underestimation of the use phase in LCAs.	Reuse stimulates new consumption.	André and Björklund (2023)
Lack of integration with macroeconomic models.	Consumption shifted to other categories.	Shinde et al. (2022)
Hypothetical data without logistics costs.	Low price generates more consumption (dress rental).	Metic et al. (2024)
Lack of empirical data.	Accessibility increases total consumption.	Mallari et al. (2024)
Little studied the sufficiency practices.	Reuse can legitimize greater consumption.	Rabiu and Jaeger-Erben (2022)
Efficiency compromises circularity.	Low-quality nutrients increase the losses.	Spiller et al. (2024)
Confusion between CE and the sharing economy.	Sharing reduces cost but increases usage.	Laurenti et al. (2019)
Excessive focus on waste.	Recycling does not replace new production.	Bartoszczuk (2023)
SHC seen as a solution generates more consumption.	Moral licensing and price effect.	Ciechelska et al. (2023)
Neglect of urban trade-offs.	Urban efficiency nullified by recreational consumption.	Fertner and Große (2016)

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 11 | Gaps in the RE in the CE.

Gaps	Authors	Forwarding proposal
Underestimation of the behavioral and cultural effects in the adoption of circular strategies	Rabiu and Jaeger-Erben (2022), André and Björklund (2023)	Include behavioral variables in models; expand studies on sufficiency practices.
Lack of analysis of the product usage phase in circular cycles.	André and Björklund (2023), Laurenti et al. (2019)	Insertion of the use phase in LCAs; empirical studies on real consumption patterns.
Absence of a systemic approach (integration between circularity, efficiency, and sufficiency).	Spiller et al. (2024), Freeman (2018)	Develop integrated models that include trade-offs and feedback.
Disconnection between the expected environmental benefits and the social and distributive impacts.	Chen (2021), Bartoszczuk (2023)	Inclusion of social and equity indicators; participatory governance.
Technocratic vision of circularity, focusing on recycling and reuse, disregarding indirect rebounds.	Mallari et al. (2024), Gheewala (2024)	Expand the approach to include consumption patterns and production systems.
Lack of specific metrics to measure the rebound effect in CE.	Freeman (2018), Shah and Zha (2025)	Development of own indicators of RE in the circular context.
Neglect of multi-scale and inter-sectoral impacts (e.g., leisure, mobility, and housing).	Fertner and Große (2016); Shinde et al. (2022)	Multi-scale and sectoral planning; integration between areas of public Policy.

degrowth. It was possible to observe that the theories can be applied in an integrative way to analyze the RE in the CE, depending on the context in which it is inserted. Table 12 presents a summary of the main findings of the study based on the theories adopted.

4.7 | Risks of the RE in the CE

The main risks of the RE in the CE, as presented in Table 13, include increases in aggregate consumption and environmental distortions, evidence of systemic and environmental risks, as well as social and economic issues.

Table 13 summarizes the main risks associated with the RE in the CE, based on selected studies between 2015 and 2025. It is organized by author, highlighting key risks such as increased aggregate consumption, moral license, environmental losses, social inequalities, and economic distortions. It is possible to verify that the RE in CE reveals some worrying patterns. Saha et al. (2024) indicate the risk of aggregate consumption and the generation of greenwashing, where companies use sustainable discourses without reducing the real impacts. In parallel, André and Björklund (2023) demonstrate that second-hand markets can intensify the consumption of products, a behavior in which the feeling of environmental contribution justifies excesses. In other words, the cultural patterns of hyperconsumerism are not being changed in circular strategies, producing unsustainable cycles.

As for the risks of social impact, studies reveal the worsening of inequalities. Rabiu and Jaeger-Erben (2022) reinforce that practices such as repair and reuse, although they reduce waste, can exclude vulnerable populations (e.g., informal workers in precarious recycling), as evidenced by Chen (2021) in studies on inequalities in India and Africa. From a macro perspective, Shah and Zha (2025) present contexts of economic instability, where circular initiatives can increase the use of fossil fuels due to the reallocation of resources to polluting sectors. These risks reveal the need for an integrated approach to prevent the CE from being a palliative, but rather a solution to planetary limits.

The risks listed in Table 13 reveal consistent patterns that reinforce the critique of the purely efficiency-based CE model, aligning with Jevons' paradox (efficiency leading to more consumption). The first pattern identified is the emphasis on increased aggregate consumption and loss of environmental benefits. This is the most recurrent pattern, appearing in at least 12 of the 18 authors (e.g., Saha et al. 2024; André and Björklund 2023; Mallari et al. 2024; Gheewala 2024). Common risks include offsetting environmental gains by increased consumption (e.g., Shinde et al. 2022), overproduction due to low prices (Metic et al. 2024), and canceling out benefits due to additional consumption (Rabiu and Jaeger-Erben 2022). The underlying pattern is that RE is seen as a systemic mechanism that transforms local gains (e.g., reuse) into negative global impacts, perpetuating linear consumption patterns.

The second pattern identified is associated with the Psychological and Behavioral Dimension. Risks such as "moral license"

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 12 | Synthesis.

Theory or theoretical background	Main findings	Authors	Frequency of occurrence
Neo-institutionalism, CMR (corporate moral responsibility), UTAUT, dynamic capabilities	Applied to the textile sector to analyze the risks of illusory decoupling and recognize systemic RE; critique of the technocratic vision and alignment with Ecological Economics and Degrowth.	Saha et al. (2024)	1
Rebound effect theory + product–user framework	It shows that user behavior can offset environmental gains; it highlights that reuse can increase total consumption, reinforcing the need for absolute decoupling.	André and Björklund (2023)	1
System dynamics (SD) + RECF (Rebound Effect Conceptual Framework)	Simulation of RE in PSS systems (e.g., dress rental); reveals how local gains generate systemic increases; used to test degrowth scenarios (consumption limits).	Metic et al. (2024)	1
Material flow analysis (MFA)	Integrates circularity, efficiency, and sufficiency in nutrient systems; reinforces that efficiency alone is not enough; connects to the concept of absolute decoupling and the degrowth agenda.	Spiller et al. (2024)	1
Social practices theory (SPT) + life cycle assessment (LCA)	It shows that circular practices often increase aggregate consumption; it highlights the need for profound social changes and sufficiency policies.	Rabiu and Jaeger-Erben (2022)	1
SD applied to prices, income and scale	Modeling price, income, and scale effects; useful for scenarios involving ecological criticism of the "illusion of efficiency"; opens space for degrowth scenarios.	Mallari et al. (2024)	1
Machine learning (random forest regression)	Used to measure RE in cooperative households; shows how efficiency gains can generate more consumption in other sectors (recoupling); aligned with the critique of ecological economics.	Shinde et al. (2022)	1

(a feeling of justification to consume more due to "sustainable" actions) and false perceptions of sustainability appear in authors such as André and Björklund (2023), Ciechelska et al. (2023), and Bertassini et al. (2022). Approximately 30% of the analyzed papers (five authors) highlight this, connecting it to greenwashing (Saha et al. 2024) and the prioritization of operational metrics over cultural changes.

The third pattern identified is associated with secondary social and economic impacts. Risks such as social inequalities, job displacement, and distortions in resource use (e.g., Saha et al. 2024; Chen 2021; Bartoszczuk 2023) are often linked to environmental risks, such as waste exports to developing countries or precarious employment in the informal recycling sector.

And the fourth concerns the pattern of scale (macro vs. micro). Micro-risks (e.g., individual behaviors in Ciechelska et al. 2023) contrast with macro-risks (e.g., loss of natural capital in Freeman 2018; scarcity of recyclable materials in Safarzynska et al. 2023). More recent studies (2023–2025) tend to emphasize systemic and global risks, reflecting a shift toward interdisciplinary analyses. Therefore, the patterns identified in the analyzed studies indicate that RE is not an isolated anomaly, but an inherent risk to CE without regulation or cultural changes, with 70% of the analyzed papers focusing on environmental/economic impacts and 30% on social/behavioral impacts. This reinforces the need for integrated approaches, such as multilevel governance, to mitigate these patterns.

While the pattern of findings is environmental/behavioral, social, and macroeconomic exceptions (Chen 2021; Safarzynska et al. 2023; Jaehn and Meissner 2022; Freeman 2018) suggest that ER is multifaceted, with potential biases in studies that ignore dimensions such as health or global climate.

4.8 | Role of Networks and Stakeholders in Mitigating the RE

Mitigating the RE in the CE depends on coordinated action between collaborative networks and multiple stakeholders. Networks are highlighted as essential structures for the coordination, dissemination of good practices, and reduction of environmental and social impacts. Freeman (2018) defends the importance of global networks to manage natural resources on a planetary scale, while Laurenti et al. (2019) propose cooperative and democratic platforms as alternatives to the dominant

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 13 | Risks of the rebound effect (RE) in the CE.

Authors	Key risks of the RE addressed
Saha et al. (2024)	Increased aggregate consumption, job displacement, greenwashing, and the dissociation between discourse and practice.
André and Björklund (2023)	Increased consumption of new products, moral licensing, and loss of expected environmental benefits.
Shinde et al. (2022)	Compensation for environmental gains with increased consumption in other categories.
Metic et al. (2024)	Overproduction due to artificial demand and excessive consumption motivated by low prices.
Mallari et al. (2024)	Aggregate increase in consumption, worse environmental performance of CE compared to linear economy.
Freeman (2018)	Possible return or amplification of the RE, loss of natural capital.
Rabiu and Jaeger-Erben (2022)	Cancelation of environmental benefits by additional consumption and circular practices justifying excess.
Spiller et al. (2024)	Reduction in systemic efficiency and increase in environmental losses due to inefficient reuse.
Laurenti et al. (2019)	Cost reduction by encouraging the overuse and increased use of consumption platforms.
Bartoszczuk (2023)	Increase in total production even with reuse/recycling, distortions in resource use.
Ciechelska et al. (2023)	Increased consumption in the SHC market (1:1.23), moral legitimization of excessive consumption.
Fertner and Große (2016)	Urban consumption trade-offs (e.g., recreational travel) and trade-offs in sustainable cities.
Bertassini et al. (2022)	Prioritization of operational metrics over profound cultural changes; poorly aligned incremental practices.
Chen (2021)	Increased social inequalities; digital exclusion; health risks in informal recycling; community conflicts.
Shah and Zha (2025)	Increased consumption of fossil fuels in contexts of economic uncertainty; compromise of environmental goals.
Gheewala (2024)	Aggregate increase in the use of natural resources; inefficient substitution of recycled materials; loss of environmental benefits.
Safarzynska et al. (2023)	Shortage of recyclable materials (scrap); rising global temperatures; reduced sustainable economic growth.
Jaehn and Meissner (2022)	"Backfire" effect in transport: total consumption higher than expected after improvements in logistics efficiency.

model of digital consumption, which often stimulates aggregate demand.

From a business perspective, Bartoszczuk (2023) argues that productive networks such as industrial symbiosis can transform waste into inputs, reducing the extraction of raw materials. At the regulatory level, Ciechelska et al. (2023) showed that initiatives such as the European Union's Green Deal exemplify how integrated policy networks can harmonize standards and raise awareness among stakeholders. On the other hand, Rabiu and Jaeger-Erben (2022) warn that poorly designed social and cultural networks can reinforce consumerist habits rather than sustainable practices. The participation of stakeholders at all levels is essential for these networks to function effectively. In the industrial sector, Saha et al. (2024) and Bartoszczuk (2023) advocate the adoption of circular design, traceability, and transparency practices. Metic et al. (2024) complement this by suggesting dynamic adjustments of inventory and prices to avoid overproduction. Consumers are called upon to adopt conscious behaviors based on reliable data, as advocated by Ciechelska et al. (2023), when they proposed the inclusion of the RE in LCAs. André and Björklund (2023) reinforce this perspective by relating educational campaigns and product durability to the reduction of moral licensing.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Government action also proves to be strategic. Shinde et al. (2022) and Mallari et al. (2024) point to the need for public policies guided by dynamic data and targeted incentives, while Freeman (2018) and Safarzynska et al. (2023) emphasize the role of international organizations in global policy coordination and in promoting fiscal solutions to foster sustainable practices. Local collaborative networks, such as the agri-food networks described by Spiller et al. (2024), demonstrate that integration between producers, governments, and communities can close production cycles without intensifying waste. Chen (2021) expands on this argument by advocating the co-creation of solutions between communities, companies, and academic institutions, promoting social justice and avoiding monopolies. Other studies reinforce this interdependent perspective. Rabiu and Jaeger-Erben (2022) suggest public support for cultural change, while Laurenti et al. (2019) propose democratic governance in digital platforms. Bertassini et al. (2022) highlight organizational cultural alignment as an essential element, and Fertner and Große (2016) highlight the role of urban managers and citizens in multi-scalar decisions. Despite their potential, these networks and partnerships face challenges. Gheewala (2024) and Shah and Zha (2025) highlight the difficulties of harmonization and regulation, especially in unstable or unequal political contexts. Still, the literature converges in stating that effective networks require transparency, such as the disclosure of RE in LCAs (Rabiu and Jaeger-Erben 2022), and inclusion mechanisms, such as subsidies for waste pickers and vulnerable populations (Spiller et al. 2024).

4.9 | Role of Innovation and Technologies in Mitigating the RE

Innovation, ranging from digital solutions to behavioral transformations, emerges as a central element in mitigating the RE in the CE. Circular business models based on rental and subscription, associated with digital traceability platforms, can reduce excessive consumption by promoting transparency and controlled access to products (Saha et al. 2024). However, André and Björklund (2023) warn that without user-centered design and behavioral innovation, such models can generate new consumption patterns, stimulated by the desire for novelty. Among the technologies analyzed, machine learning (Shinde et al. 2022) and SD (Metic et al. 2024) allow simulating scenarios and predicting consumption patterns and environmental impacts, offering support for adaptive decision-making. Saha et al. (2024) and Gheewala (2024) highlight the role of Industry 4.0 technologies (e.g., blockchain, IoT, automated material tracking, and separation) in the efficient management of circular supply chains. However, such innovations can intensify inequalities if they are not accompanied by digital inclusion strategies (Chen 2021).

In turn, Rabiu and Jaeger-Erben (2022) highlight the role of social innovation, which transforms the cultural relationship with consumption and thus reduces the rebound. This approach is reinforced by educational initiatives (Ciechelska et al. 2023), which are essential to avoid compensatory purchases motivated by a perception of moral license.

The adoption of digital technologies, such as sharing platforms (Airbnb and Uber), despite facilitating circular practices, can generate an increase in aggregate consumption in the absence of regulation (Laurenti et al. 2019). In the agri-food sector, Spiller et al. (2024) highlight the potential of technologies to reduce waste but warn of imbalances generated by less efficient substitutions. To monitor such effects, Bartoszczuk (2023) proposed real-time digital indicators, while Rabiu and Jaeger-Erben (2022) advocated the integration of technologies with life cycle analysis (LCA) as a guide for public policies. In short, the effectiveness of technological innovations depends on their alignment with socio-environmental criteria, inclusive policies, and cultural changes. Freeman (2018) highlights the need for technological efficiency combined with institutional governance; Laurenti et al. (2019) emphasize the importance of social justice and platform regulation to prevent increased consumption; and Bertassini et al. (2022) reinforce that the transformation of consumption patterns requires profound cultural changes and alignment between innovation and organizational values.

4.10 | The Role of Governance in Mitigating the RE

The analysis of the articles demonstrates an emphasis on multilevel governance as a mechanism to mitigate the RE in the CE. In this sense, it is possible to say that governance plays an important role in mitigating the RE in the CE, since it is through governance that institutional, regulatory, and collaborative mechanisms are defined that guide the behavior of stakeholders. Saha et al. (2024) highlight the need for robust multilevel governance (companies, government, and NGOs) to avoid decoupling. In a complementary way, André and Björklund (2023) emphasize the need for multilevel and sectoral governance that promotes transparency, encourages effective regulation, and aligns incentives to prevent circular strategies, such as reuse and recycling, from generating contrary effects. These studies demonstrate that robust governance is necessary to prevent a decoupling between intention and practice. In the same vein, Metic et al. (2024) highlight the involvement of multiple actors in data-based governance and simulations for adaptive decision-making, highlighting the importance of coordinated actions to mitigate the RE, such as adjustments in recovery capacity and consumption behavior. Mallari et al. (2024) argue that the role of governance goes beyond standardization and should be systemic and articulated with policies and user behavior.

In turn, Bartoszczuk (2023) views governance as essential to advancing the CE, but warns about the need for monitoring to avoid the RE, thus defending governance based on regulations, socio-environmental standards, and metrics (such as the BS 8001:2017 standard). In contrast, Rabiu and Jaeger-Erben (2022) highlight the importance of participatory governance for the institutional and cultural transformation of consumption practices, while Chen (2021) complements this view by presenting social inclusion and transparency to promote justice and mitigate inequalities. These approaches reflect the different views on how to operationalize governance: presenting, on the one hand, the top-down model (via regulation) and, on the other, the bottom-up (participatory) framework. Furthermore, Freeman (2018) broadens the scope by highlighting international and intergovernmental governance

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

for managing externalities and natural capital. Shah and Zha (2025) argue that macroeconomic governance, based on political stability and environmental regulation, is effective in moderating the energy RE.

4.11 | Discussion

The study confirms that the classic logic of the RE—efficiency gains leading to increased consumption—remains valid in different CE strategies (reuse, recycling, PSS, and sharing). This shows that, despite methodological innovation (machine learning, SD, and LCA), the conceptual foundations still converge on the same criticism: efficiency does not guarantee sustainability. Many studies still operate with a technocratic perspective, focused on material/energy metrics, underestimating cultural and behavioral aspects that intensify RE.

The use of SD, machine learning, MFA, and SPT broadens the understanding of RE by integrating economic, social, and psychological dimensions. Although methodologically sound, many studies remain sectoral and fragmented, hindering integrative multiscale models that articulate everything from the consumer level to macroeconomic impacts. Furthermore, evidence that circular practices (e.g., secondhand clothing and rental) legitimize increased consumption through moral licensing shows that technological change without cultural change can reinforce linear consumption. Another relevant finding is that few studies offer effective solutions to this dilemma, such as education for sufficiency or consumption restriction models, which highlights a gap for future research. Although present, the connection with Ecological Economics and Degrowth needs to advance to propose robust indicators of absolute, not just relative, decoupling.

The empirical analysis also indicates that circular practices, such as clothing rental (Metic et al. 2024) or second-hand purchases (Ciechelska et al. 2023), have produced substantial behavioral rebounds, with emphasis on the phenomenon of moral license, where the perception of sustainability legitimizes excessive consumption. In this context, CE goes from being a promising solution to a systemic palliative—an injection of sustainability that masks structural problems without changing them.

Governance, in turn, emerges as a key element for mitigating rebounds. Studies such as those by Saha et al. (2024) and Bartoszczuk (2023) demonstrate that the absence of integrated public policies, regulation based on socio-environmental metrics, and stakeholder involvement tends to amplify the negative effects of the rebound. Multilevel, transparent, and data-driven governance (Metic et al. 2024; Chen 2021) thus becomes a necessary condition for effective and socially just circular transitions.

In addition, the results corroborate the argument that technological innovation alone is insufficient. Digital platforms, blockchain, rental models, and IoT, if not combined with redistributive policies and education for sufficiency, can generate new cycles of overconsumption, especially in peripheral economies (Rabiu and Jaeger-Erben 2022; Gheewala 2024). Therefore, innovation needs to be also institutional, cultural, and normative—connected to an agenda of structural transformation.

In this sense, the concept of regenerative CE appears as a powerful alternative, in line with the critique of degrowth and the well-being economy. This requires that CE not only minimize impacts but also rethink the value, use, and role of resources in production systems. The integration of circularity, efficiency, and sufficiency, as proposed by Spiller et al. (2024), offers a promising path, still little explored, but with real disruptive potential.

The study forms a multifactorial integrative framework, systematically bringing together circular strategies, types of renewable energy, applied theories, and mitigation instruments. The study highlights that, without multiscale governance and education for sufficiency, renewable energy can reproduce unsustainable patterns of the linear model. It articulates renewable energy with degrowth, ecological economics, social practices, and dynamic capabilities, generating a more plural and critical theoretical framework.

The finding that "circular consumption" can increase total consumption highlights the need for cultural changes beyond technological innovation. This implies public policies and educational campaigns that encourage sufficiency, away from the logic of hyperconsumption masquerading as sustainable.

The research shows that circularity without adequate metrics and without production limits can result in greenwashing and a loss of credibility in ESG. It suggests the adoption of hybrid indicators (quantitative and qualitative) that measure both environmental and behavioral impacts to avoid illusions of sustainability. It reinforces the urgency of multilevel governance, with regulatory instruments (taxation of virgin resources, production limits, regulation of digital platforms) combined with educational and participatory mechanisms. It implies recognizing RE as a systemic risk that can neutralize the advances of CE, requiring public policies that combine efficiency and sufficiency.

The study opens the door to multiscale empirical research, integrating everything from everyday practices to global impacts. It implies the construction of integrative models for measuring RE, with the potential to guide both public policies and business strategies. Therefore, the study offers an original contribution by demonstrating that the CE, if guided solely by efficiency, risks reinforcing linear consumption. Its findings alert society to the danger of the "circularity myth" and indicate that only a combination of robust governance, sufficiency education, and critical theories can ensure that the CE advances in a fair and sustainable manner.

Finally, in Figure 2, we present the proposal of a framework that proposes a systemic approach to CE, considering not only circular strategies but also risk factors, types of RE—when efficiency gains are offset by increased consumption—and mitigation instruments. The framework comes from the original findings of the study and the gaps identified in the field that demand the advancement of critical reflection, analysis, and effective action. It is structured around six axes: circular strategies (inputs), risk factors, types of RE, mitigation instruments, expected results, and practical indicators to monitor and mitigate the RE.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

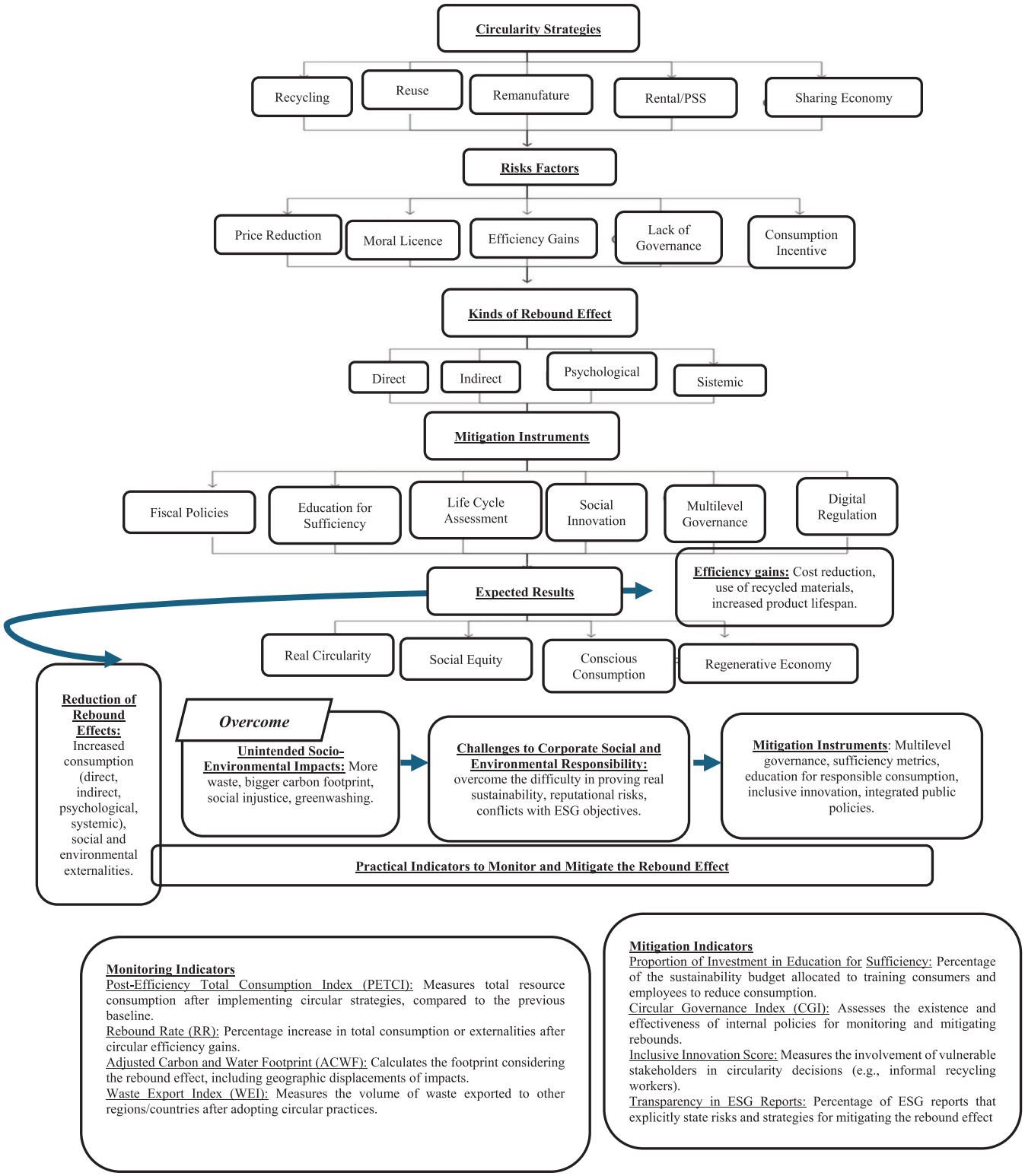

FIGURE 2 | Framework summarizing the original study findings.

The framework (Figure 2) aims to shift from a technocratic CE (focused solely on efficiency) to a regenerative vision that balances technical practices (such as LCAs), behavioral changes (conscious consumption), and governance mechanisms (multilevel regulations). The RE in the CE represents a central challenge for corporate social and environmental responsibility. Recognizing, measuring, and mitigating this phenomenon is essential to ensure that CE effectively contributes to sustainability and does not become just another greenwashing strategy. The proposed framework and suggested indicators offer practical and theoretical tools for companies and researchers to move toward a genuinely responsible and transformative circularity.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

5 | Final Remarks

This study conducted an SLR of 75 articles published between 2015 and 2025 to critically examine how contemporary scholarship addresses the RE within CE strategies, revealing its potential to undermine anticipated environmental, economic, and social gains. The analysis underscores a persistent pattern: while CE promises resource decoupling through practices like reuse, recycling, remanufacturing, and PSS, unchecked RE—manifesting as direct (e.g., increased consumption from lower costs), indirect (e.g., income reallocation to other sectors), psychological (e.g., moral licensing justifying overconsumption), and systemic (e.g., economy-wide demand surges)—often neutralizes these benefits, as evidenced by sectoral cases in textiles, energy, transportation, and construction. Recent advancements, such as the 2024 Circularity Gap Report highlighting Europe's overreliance on virgin materials (87%) and a 2025 study on Dutch building renovations demonstrating that refurbishment can inflate material use by 21%–25% and costs by 2%–6% while yielding minimal emission reductions (up to 0.3%), further illustrate how efficiency gains can cascade into amplified resource extraction and emissions, particularly without aligned energy transitions or enhanced circularity measures that offset up to 89% of material footprints.

Theoretically, this research advances the field by integrating ecological economics' critique of relative decoupling with social practice theory, SD, and sufficiency frameworks, proposing a regenerative CE model that transcends technocratic efficiency. This alignment is substantiated by the SLR's categorization of RE mechanisms (e.g., price effects in second-hand markets offsetting 100% of savings in smartphone reuse) and mitigation tools (e.g., multilevel governance and LCA-integrated metrics).

Practically, the findings equip policymakers with evidence-based strategies, such as virgin resource taxes and education for sufficiency to counteract behavioral rebounds, while advising corporate managers to embed RE monitoring in ESG reporting to avoid greenwashing—insights validated by the review's identification of gaps in governance and cultural alignment. For educators, incorporating rebound-aware curricula fosters planetary citizenship, addressing the SLR's noted absence of sufficiency practices in current CE implementations.

Limitations include restricted access to some articles and a predominance of Global North perspectives, potentially overlooking Global South dynamics like informal recycling precarity. Future research should empirically validate RE in emerging chains (e.g., bioeconomy), develop multi-scale indicators integrating recent methods like computable general equilibrium models and stock-flow consistent modeling from 2024 reviews, and explore post-2025 innovations, such as AI-enhanced traceability to mitigate cascading rebounds. Ultimately, CE's transformative potential hinges on evolving from palliative efficiency to holistic regeneration, demanding interdisciplinary agendas that confront planetary limits head-on for equitable sustainability. Also, consider disaggregating the analysis of the RE by country or region.

For future research, we recommend an empirical deepening of the mechanisms of rebound in specific production chains (e.g., textiles, food, and construction), as well as the development of multi-scale indicators that integrate environmental, social, and cultural aspects. Longitudinal and participatory studies are also suggested to map the real effects of circular policies on consumption patterns in different territories. In short, the advancement of CE as a robust solution requires a paradigmatic break from an instrumental and technocratic circularity to a reflective, fair circularity aligned with planetary limits. Only then will it be possible to transition from a potentially regenerative model to an actual sustainable reality. Future studies can also explore the relationship between the CE and governance.

Acknowledgments

This study was supported by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), Fapesc - Fundaçãod e Ampara e Pesquisa do Estado de Santa Catarina, and the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES). The Article Processing Charge for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614).

References

André, H., and A. Björklund. 2023. "A Framework to Open the Black Box of the Use Phase in Circular Economy Life Cycle Assessments: The Case of Shell Jacket Reuse." *Journal of Industrial Ecology* 27, no. 4: 1137–1150. https://doi.org/10.1111/jiec.13408.

Arpin, M. L., S. H. Leclerc, and G. Lonca. 2024. "The Circular Economy (CE) Rebound as a Paradox of Knowledge: Forecasting the Future of the CE–IoT Nexus Through the Global E-Waste Crisis." *Sustainability* 16, no. 15: 6364.

Bączyk, M., V. Tunn, E. Worrell, and B. Corona. 2024. "Consumer Behavior in Circular Business Models: Unveiling Conservation and Rebound Effects." *Sustainable Production and Consumption* 52: 283–298. https://doi.org/10.1016/j.spc.2024.10.022.

Bardin, L. 2015. *Análise de conteúdo*, 70. Edições.

Bartoszczuk, P. 2023. "Circular Economy and Its Restriction." *Economics and Environment* 86, no. 3: 1–14. https://doi.org/10.34659/eis.2023.86.3.650.

Bauer, T., P. Zwolinski, N. Nasr, and G. Mandil. 2020. "Characterization of Circular Strategies to Better Design Circular Industrial Systems." *Journal of Remanufacturing* 10, no. 3: 161–176.

Bauwens, T. 2021. "Are the Circular Economy and Economic Growth Compatible? A Case for Post-Growth Circularity." *Resources, Conservation and Recycling* 175: 105852.

Bertassini, A. C., L. D. D. R. Calache, L. C. R. Carpinetti, A. R. Ometto, and M. C. Gerolamo. 2022. "CE-Oriented Culture Readiness: An Assessment Approach Based on Maturity Models and Fuzzy Set Theories." *Sustainable Production and Consumption* 31: 615–629. https://doi.org/10.1016/j.spc.2022.03.018.

Beyeler, L., and M. Jaeger-Erben. 2022. "How to Make More of Less: Characteristics of Sufficiency in Business Practices." *Frontiers in Sustainability* 3: 949710.

Birat, J. P. 2015. "Life Cycle Assessment, Resource Efficiency and Recycling." *Metallurgical Research & Technology* 112, no. 2: 206.

Bocken, N., K. Miller, and S. Evans. 2016. "Assessing the Environmental Impact of New Circular Business Models." *In Proceedings of the "New Business Models"—Exploring a Changing View on Organizing Value Creation*, Toulouse, France, vol. 1, 16–17. https://www.researchgate.net/publication/305264490.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Bocken, N. M., L. Niessen, and S. W. Short. 2022. "The Sufficiency-Based Circular Economy—An Analysis of 150 Companies." *Frontiers in Sustainability* 3: 899289.

Bubinek, R., U. Knaack, and C. Cimpan. 2025. "Reuse of Consumer Products: Climate Account and Rebound Effects Potential." *Sustainable Production and Consumption* 54: 190–201.

Castro, C. G., A. H. Trevisan, D. C. Pigosso, and J. Mascarenhas. 2022. "The Rebound Effect of Circular Economy: Definitions, Mechanisms and a Research Agenda." *Journal of Cleaner Production* 345: 131136.

Chatzistamoulou, N., and P. Koundouri. 2024. "Is Green Transition in Europe Fostered by Energy and Environmental Efficiency Feedback Loops? The Role of Eco-Innovation, Renewable Energy and Green Taxation." *Environmental and Resource Economics* 87, no. 6: 1445–1472.

Chatzistamoulou, N., and E. Tyllianakis. 2022. "Green Growth & Sustainability Transition Through Information. Are the Greener Better Informed? Evidence From European SMEs." *Journal of Environmental Management* 306: 114457. https://doi.org/10.1016/j.jenvman.2022.114457.

Chen, C. W. 2021. "Clarifying Rebound Effects of the Circular Economy in the Context of Sustainable Cities." *Sustainable Cities and Society* 66: 102622. https://doi.org/10.1016/j.scs.2020.102622.

Ciechelska, A., A. Matel, T. Poskrobko, and E. Sidorczuk-Pietraszko. 2023. "Circular Economy Rebound Effect in the Context of Second-Hand Clothing Consumption." *Ekonomia i Środowisko* 87, no. 4: 1–20. https://doi.org/10.34659/eis.2023.87.4.635.

Corvellec, H., A. F. Stowell, and N. Johansson. 2022. "Critiques of the Circular Economy." *Journal of Industrial Ecology* 26, no. 2: 421–432. https://doi.org/10.1111/jiec.13187.

Csutora, M., G. Harangozo, and C. Szigeti. 2022. "Sustainable Business Models—Crisis and Rebound Based on Hungarian Research Experience." *Resources* 11, no. 12: 107.

Das, A., J. Konietzko, N. Bocken, and M. Dijk. 2023. "The Circular Rebound Tool: A Tool to Move Companies Towards More Sustainable Circular Business Models." *Resources, Conservation & Recycling Advances* 20: 200185.

Fan, F., K. Zhang, S. Dai, and X. Wang. 2023. "Decoupling Analysis and Rebound Effect Between China's Urban Innovation Capability and Resource Consumption." *Technology Analysis & Strategic Management* 35, no. 4: 478–492.

Ferrante, M., M. Vitti, S. Digiesi, and C. Sassanelli. 2023. "Investigating the Circular Economy's Rebound Effect in Manufacturing: A Systematic Literature Review." In *Proceedings of the Summer School Francesco Turco Genoa*.

Ferrante, M., M. Vitti, F. Facchini, and C. Sassanelli. 2024. "Mapping the Relations Between the Circular Economy Rebound Effects Dimensions: A Systematic Literature Review." *Journal of Cleaner Production* 456: 142399.

Fertner, C., and J. Große. 2016. "Compact and Resource Efficient Cities? Synergies and Trade-Offs in European Cities." *European Spatial Research and Policy* 23, no. 1: 65–79. https://doi.org/10.1515/esrp-2016-0004.

Figge, F., and A. S. Thorpe. 2019. "The Symbiotic Rebound Effect in the Circular Economy." *Ecological Economics* 163: 61–69.

Flachenecker, F. 2018. "The Causal Impact of Material Productivity on Macroeconomic Competitiveness in the European Union." *Environmental Economics and Policy Studies* 20, no. 1: 17–46.

Flores, J., M. Cavique, and J. Seixas. 2022. "Energy Sustainability—Rebounds Revisited Using Axiomatic Design." *Sustainability* 14, no. 11: 6737.

Font Vivanco, D., J. Freire-González, R. Galvin, et al. 2022. "Rebound Effect and Sustainability Science: A Review." *Journal of Industrial Ecology* 26, no. 4: 1543–1563.

Font Vivanco, D., S. Sala, and W. McDowall. 2018. "Roadmap to Rebound: How to Address Rebound Effects From Resource Efficiency Policy." *Sustainability* 10, no. 6: 2009.

Fraser, M., Á. Conde, and L. Haigh. 2024. "Circularity Gap Report 2024." https://coilink.org/20.500.12592/fbg7g7x.

Freeman, R. 2018. "A Theory on the Future of the Rebound Effect in a Resource-Constrained World." *Frontiers in Energy Research* 6: 81. https://doi.org/10.3389/fenrg.2018.00081.

Geissdoerfer, M., P. Savaget, N. M. Bocken, and E. J. Hultink. 2017. "The Circular Economy—A New Sustainability Paradigm?" *Journal of Cleaner Production* 143: 757–768. https://doi.org/10.1016/j.jclepro.2016.12.048.

Geng, Y., T. Fujita, R. Bleischwitz, A. Chiu, and J. Sarkis. 2019. "Accelerating the Transition to Equitable, Sustainable, and Livable Cities: Toward Post-Fossil Carbon Societies." *Journal of Cleaner Production* 239: 118020. https://doi.org/10.1016/j.jclepro.2019.118020.

Genovese, A., B. H. Lowe, M. Bimpizas-Pinis, and V. G. Ram. 2024. "Planning for a Future Free From Rebound Effects." *Futures* 164: 103479.

Gheewala, S. H. 2024. "Material Recycling in a Circular Economy—A Systems View." *Wiley Interdisciplinary Reviews: Energy and Environment* 13, no. 1: e503. https://doi.org/10.1002/wene.503.

Godinot, O., J. Jouan, T. Nesme, and M. Carof. 2024. "Evidence of a Rebound Effect in Agriculture: Crop-Livestock Reconnection Beyond the Farm Gate Does Not Always Lead to More Sustainable Nitrogen Management." *Agricultural Systems* 221: 104137. https://doi.org/10.1016/j.agsy.2024.104137.

Gossart, C. 2014. "Rebound Effects and ICT: A Review of the Literature." *ICT Innovations for Sustainability*: 435–448.

Grădinaru, G. I., and A. C. Maricuț. 2022. "From the Rebound Effect to the Perspective of Circular Economy: A Structure Changes Analysis Among Eu Countries." *Economic Computation & Economic Cybernetics Studies & Research* 56, no. 1.

Guzzo, D., and D. C. Pigosso. 2024. "Identifying Rebound Effects in Product-Service Systems: Actors, Mechanisms, Triggers and Drivers." *Proceedings of the Design Society* 4: 1279–1288.

Henry, M., D. Schraven, N. Bocken, K. Frenken, M. Hekkert, and J. Kirchherr. 2021. "The Battle of the Buzzwords: A Comparative Review of the Circular Economy and the Sharing Economy Concepts." *Environmental Innovation and Societal Transitions* 38: 1–21.

Huang, C., and B. Lin. 2025. "The Impact of Digital Economy on Energy Rebound Effect in China: A Stochastic Energy Demand Frontier Approach." *Energy Policy* 196: 114418.

Jaehn, F., and F. Meissner. 2022. "The Rebound Effect in Transportation." *Omega* 108: 102563. https://doi.org/10.1016/j.omega.2021.102563.

Jevons, W. S. 1865. *The Coal Question: An Inquiry Concerning the Progress of the Nation, and the Probable Exhaustion of Our Coal Mines*. Macmillan.

Kirchherr, J. 2022. "Circular Economy and Growth: A Critical Review of "Post-Growth" Circularity and a Plea for a Circular Economy That Grows." *Resources, Conservation and Recycling* 179: 106033.

Kjaer, L. L., D. C. Pigosso, T. C. McAloone, and M. Birkved. 2018. "Guidelines for Evaluating the Environmental Performance of Product/Service-Systems Through Life Cycle Assessment." *Journal of Cleaner Production* 190: 666–678.

Kjaer, L. L., D. C. Pigosso, M. Niero, N. M. Bech, and T. C. McAloone. 2019. "Product/Service-Systems for a Circular Economy: The Route to Decoupling Economic Growth From Resource Consumption?" *Journal of Industrial Ecology* 23, no. 1: 22–35.

Koide, R., S. Murakami, and K. Nansai. 2022. "Prioritising Low-Risk and High-Potential Circular Economy Strategies for Decarbonisation:

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

A Meta-Analysis on Consumer-Oriented Product-Service Systems." *Renewable and Sustainable Energy Reviews* 155: 111858.

Konash, A., and N. Nasr. 2022. "The Circular Economy and Resource Use Reduction: A Case Study of Long-Term Resource Efficiency Measures in a Medium Manufacturing Company." *Cleaner Production Letters* 3: 100025.

Korhonen, J., C. Nuur, A. Feldmann, and S. E. Birkie. 2018. "Circular Economy as an Essentially Contested Concept." *Journal of Cleaner Production* 175: 544–552. https://doi.org/10.1016/j.jclepro.2017.12.111.

Lahcen, B., J. Eyckmans, and J. Brusselaers. 2024. "Linking Methodologies to Assess Climate Impacts and Circular Economy Strategies Along Supply Chains." *Sustainable Production and Consumption* 47: 544–556.

Laurenti, R., J. Singh, J. M. Cotrim, M. Toni, and R. Sinha. 2019. "Characterizing the Sharing Economy State of the Research: A Systematic Map." *Sustainability* 11, no. 20: 5729. https://doi.org/10.3390/su11205729.

Levänen, J., V. Uusitalo, A. Härri, E. Kareinen, and L. Linnanen. 2021. "Innovative Recycling or Extended Use? Comparing the Global Warming Potential of Different Ownership and End-Of-Life Scenarios for Textiles." *Environmental Research Letters* 16, no. 5: 054069.

Liu, X., S. Wang, Y. Li, et al. 2022. "RETRACTED: Recycling in Textile Sector: A New Circular Economy Approach Towards Ecology and Environmental Sustainability." *Frontiers in Environmental Science* 10: 929710.

Lowe, B. H., M. Bimpizas-Pinis, P. Zerbino, and A. Genovese. 2024. "Methods to Estimate the Circular Economy Rebound Effect: A Review." *Journal of Cleaner Production* 443: 141063. https://doi.org/10.1016/j.jclepro.2024.141063.

Makov, T., and D. Font Vivanco. 2018. "Does the Circular Economy Grow the Pie? The Case of Rebound Effects From Smartphone Reuse." *Frontiers in Energy Research* 6: 39.

Mallari, C. B., R. Li, and J. L. San Juan. 2024. "A System Dynamics Model for Analyzing the Circular Economy Rebound." *Chemical Engineering Transactions* 114: 337–342. https://doi.org/10.3303/CET24114057.

Mata, É., J. M. Kihila, J. Wanemark, et al. 2022. "Non-Technological and Behavioral Options for Decarbonizing Buildings–a Review of Global Topics, Trends, Gaps, and Potentials." *Sustainable Production and Consumption* 29: 529–545.

Metic, J., D. Guzzo, B. Kopainsky, T. C. McAloone, and D. C. Pigosso. 2024. "A Simulation-Based Approach for Investigating the Dynamics of Rebound Effects in the Circular Economy: A Case of Use-Oriented Product/Service System." *Journal of Environmental Management* 365: 121627. https://doi.org/10.1016/j.jenvman.2024.121627.

Metic, J., and D. C. Pigosso. 2022. "Research Avenues for Uncovering the Rebound Effects of the Circular Economy: A Systematic Literature Review." *Journal of Cleaner Production* 368: 133133.

Morimoto, S., H. Kuroki, H. Narita, and A. Ishigaki. 2021. "Scenario Assessment of Neodymium Recycling in Japan Based on Substance Flow Analysis and Future Demand Forecast." *Journal of Material Cycles and Waste Management* 23, no. 6: 2120–2132.

Murray, A., K. Skene, and K. Haynes. 2017. "The Circular Economy: An Interdisciplinary Exploration of the Concept and Application in a Global Context." *Journal of Business Ethics* 140: 369–380. https://doi.org/10.1007/s10551-015-2693-2.

Niehoff, S., and G. Beier. 2018. "Industrie 4.0 and a Sustainable Development: A Short Study on the Perception and Expectations of Experts in Germany." *International Journal of Innovation and Sustainable Development* 12, no. 3: 360–374.

Niero, M., C. L. Jensen, C. F. Fratini, J. Dorland, M. S. Jørgensen, and S. Georg. 2021. "Is Life Cycle Assessment Enough to Address Unintended Side Effects From Circular Economy Initiatives?" *Journal of Industrial Ecology* 25, no. 5: 1111–1120.

Nußholz, J., S. Çetin, L. Eberhardt, C. De Wolf, and N. Bocken. 2023. "From Circular Strategies to Actions: 65 European Circular Building Cases and Their Decarbonisation Potential." *Resources, Conservation & Recycling Advances* 17: 200130.

Onat, N. C., J. Mandouri, M. Kucukvar, et al. 2023. "Rebound Effects Undermine Carbon Footprint Reduction Potential of Autonomous Electric Vehicles." *Nature Communications* 14, no. 1: 6258.

Ottelin, J., H. Cetinay, and P. Behrens. 2020. "Rebound Effects May Jeopardize the Resource Savings of Circular Consumption: Evidence From Household Material Footprints." *Environmental Research Letters* 15, no. 10: 104044.

Pádua Pieroni, M., F. Blomsma, T. C. McAloone, and D. C. Pigosso. 2018. "Enabling Circular Strategies With Different Types of Product/Service-Systems." *Procedia CIRP* 73: 179–184.

Piscicelli, L. 2023. "The Sustainability Impact of a Digital Circular Economy." *Current Opinion in Environmental Sustainability* 61: 101251.

Pohl, J., L. M. Hilty, and M. Finkbeiner. 2019. "How LCA Contributes to the Environmental Assessment of Higher Order Effects of ICT Application: A Review of Different Approaches." *Journal of Cleaner Production* 219: 698–712.

Prieto-Sandoval, V., C. Jaca, J. Santos, R. J. Baumgartner, and M. Ormazabal. 2019. "Key Strategies, Resources, and Capabilities for Implementing Circular Economy in Industrial Small and Medium Enterprises." *Corporate Social Responsibility and Environmental Management* 26, no. 6: 1473–1484. https://doi.org/10.1002/csr.1761.

Provensi, T., and S. Sehnem. 2025. "Toward Circular Transition: Exploring the Role of Circular Start-Ups." *Sustainable Production and Consumption* 57: 46–60. https://doi.org/10.1016/j.spc.2025.05.009.

Rabiu, M. K., and M. Jaeger-Erben. 2022. "Appropriation and Routinisation of Circular Consumer Practices: A Review of Current Knowledge in the Circular Economy Literature." *Cleaner and Responsible Consumption* 7: 100081. https://doi.org/10.1016/j.clrc.2022.100081.

Roa, D., and K. E. Rosendahl. 2023. "Policies for Material Circularity: The Case of Lithium." *Circular Economy and Sustainability* 3, no. 1: 373–405.

Safarzynska, K., L. Di Domenico, and M. Raberto. 2023. "The Circular Economy Mitigates the Material Rebound due to Investments in Renewable Energy." *Journal of Cleaner Production* 402: 136753. https://doi.org/10.1016/j.jclepro.2023.136753.

Saha, K., P. K. Dey, and V. Kumar. 2024. "A Comprehensive Review of Circular Economy Research in the Textile and Clothing Industry." *Journal of Cleaner Production* 444: 141252. https://doi.org/10.1016/j.jclepro.2024.141252.

Salvador, R., M. V. Barros, M. Donner, P. Brito, A. Halog, and A. C. De Francisco. 2022. "How to Advance Regional Circular Bioeconomy Systems? Identifying Barriers, Challenges, Drivers, and Opportunities." *Sustainable Production and Consumption* 32: 248–269.

Salvador, R., F. N. Puglieri, A. Halog, F. G. de Andrade, C. M. Piekarski, and A. C. De Francisco. 2021. "Key Aspects for Designing Business Models for a Circular Bioeconomy." *Journal of Cleaner Production* 278: 124341.

Samani, P. 2025. "Replaying the Circular Economy Orchestra: The Echoes of R and the Symphony of I." *Journal of Integrative Environmental Sciences* 22, no. 1: 2440156.

Santarius, T., H. J. Walnum, and C. Aall. 2018. "From Unidisciplinary to Multidisciplinary Rebound Research: Lessons Learned for Comprehensive Climate and Energy Policies." *Frontiers in Energy Research* 6: 104.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Saunders, H. D. 1992. "The Khazzoom-Brookes Postulate and Neoclassical Growth." *Energy Journal* 13, no. 4: 131–148. https://doi.org/10.5547/ISSN0195-6574-EJ-Vol13-No4-7.

Scarpellini, S. 2022. "Social Impacts of a Circular Business Model: An Approach From a Sustainability Accounting and Reporting Perspective." *Corporate Social Responsibility and Environmental Management* 29, no. 3: 646–656. https://doi.org/10.1002/csr.2226.

Schultz, F. C. 2022. "The Circular Economy and Economic Growth—An Irreconcilable Tradeoff?" *Resources, Conservation and Recycling* 183: 106351.

Schultz, F. C., and I. Pies. 2024. "The Circular Economy Growth Machine: A Critical Perspective on "Post-Growth" and "Pro-Growth" Circularity Approaches." *Journal of Industrial Ecology* 28, no. 1: 17–24.

Schultz, F. C., V. Valentinov, R. J. Reinhardt, and I. Pies. 2024. "The Circular Economy Rebound Effect: Reconceptualizing Rebound Approaches and Mitigation Opportunities From an Ordonomic Perspective." *Journal of Industrial Ecology* 28, no. 3: 374–385.

Sehnem, S., I. Junges, S. V. Soares, and A. Maganda. 2025. "Reconciling the Efficiency of Industry 4.0 and Circular Economy With the Human Empowerment of Lean: Unraveling the Challenges and Opportunities of Integration." *Social Responsibility Journal* 21, no. 3: 429–450.

Shah, A. A., and D. Zha. 2025. "Economy-Wide Estimates of the Energy Rebound Effect in BRICS: The Role of Environmental Regulations and Economic Policy Uncertainty." *Energy* 320: 135276. https://doi.org/10.1016/j.energy.2025.135276.

Sheng, X., X. Zhang, and X. Zhou. 2023. "Show Me the Impact: Communicating "Behavioral Impact Message" to Promote Pro-Environmental Consumer Behavior." *Sustainable Production and Consumption* 35: 709–723.

Shinde, R., A. Froemelt, A. Kim, and S. Hellweg. 2022. "A Novel Machine Learning Approach for Evaluating Rebounds-Associated Environmental Footprint of Households and Application to Cooperative Housing." *Journal of Environmental Management* 304: 114205. https://doi.org/10.1016/j.jenvman.2021.114205.

Siderius, T., and K. Poldner. 2021. "Reconsidering the Circular Economy Rebound Effect: Propositions From a Case Study of the Dutch Circular Textile Valley." *Journal of Cleaner Production* 293: 125996.

Skene, K. R. 2018. "Circles, Spirals, Pyramids and Cubes: Why the Circular Economy Cannot Work." *Sustainability Science* 13, no. 2: 479–492. https://doi.org/10.1007/s11625-017-0443-3.

Sorrell, S. 2009. "Jevons' Paradox Revisited: The Evidence for Backfire From Improved Energy Efficiency." *Energy Policy* 37, no. 4: 1456–1469. https://doi.org/10.1016/j.enpol.2008.12.003.

Sourabh, S., B. G. Menon, and B. Mahanty. 2024. "Econometric Analysis of Circular Economy Co-Flow Process in Metal Industry." *Quality & Quantity* 58, no. 2: 1583–1602.

Spiller, M., R. Vingerhoets, S. E. Vlaeminck, F. Wichern, and A. Papangelou. 2024. "Beyond Circularity! Integration of Circularity, Efficiency, and Sufficiency for Nutrient Management in Agri-Food Systems." *Nutrient Cycling in Agroecosystems* 129, no. 3: 287–297. https://doi.org/10.1007/s10705-024-10339-8.

Tranfield, D., D. Denyer, and P. Smart. 2003. "Toward a Methodology for Developing Evidence-Informed Management Knowledge by Means of Systematic Review." *British Journal of Management* 14, no. 3: 207–222. https://doi.org/10.1111/1467-8551.00375.

Tsuji, K., K. Kurisu, J. Nakatani, and Y. Moriguchi. 2020. "Evaluation of Environmental Impact of Car Sharing in Consideration of Uncertainty of Influential Variables." *International Journal of Automation Technology* 14, no. 6: 975–983.

Tutore, I., A. Parmentola, M. C. di Fiore, and F. Calza. 2024. "A Conceptual Model of Artificial Intelligence Effects on Circular Economy Actions." *Corporate Social Responsibility and Environmental Management* 31, no. 5: 4772–4782. https://doi.org/10.1002/csr.2827.

Vegter, D., J. van Hillegersberg, and M. Olthaar. 2023. "Performance Measurement System for Circular Supply Chain Management." *Sustainable Production and Consumption* 36: 171–183.

Wallenborn, G. 2018. "Rebounds Are Structural Effects of Infrastructures and Markets." *Frontiers in Energy Research* 6: 99.

Warmington-Lundström, J., and R. Laurenti. 2020. "Reviewing Circular Economy Rebound Effects: The Case of Online Peer-To-Peer Boat Sharing." *Resources, Conservation & Recycling: X* 5: 100028.

Wiprächtiger, M., M. Rapp, S. Hellweg, R. Shinde, and M. Haupt. 2022. "Turning Trash Into Treasure: An Approach to the Environmental Assessment of Waste Prevention and Its Application to Clothing and Furniture in Switzerland." *Journal of Industrial Ecology* 26, no. 4: 1389–1405.

Yin, R. 2016. *Pesquisa qualitativa do início ao fim*. Penso.

Zerbino, P. 2022. "How to Manage the Circular Economy Rebound Effect: A Proposal for Contingency-Based Guidelines." *Journal of Cleaner Production* 378: 134584. https://doi.org/10.1016/j.jclepro.2022.134584.

Zink, T., and R. Geyer. 2017. "Circular Economy Rebound." *Journal of Industrial Ecology* 21, no. 3: 593–602.

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix A

The exclusion criteria were: areas other than Business Management and Accounting; Papers published in languages other than English, Portuguese, and Spanish; Papers published before 2015; Duplicate Papers; Papers not available in full, and Papers outside the scope of the research.

The inclusion criteria used were: Business Management and Accounting; English, Portuguese, and Spanish; Article and Review; Year: 2015–2025; The word "rebound effect" must appear in the title, abstract, or keywords; the article must mention types of rebound effect and the impacts generated; and the article must mention sustainable circular business models.

TABLE A1 | Bibliographic portfolio analyzed.

No.	References	Country
1	André, H. 2024. "Opening the Black Box of the Use Phase in Circular Economy Life Cycle Assessments: Environmental Performance of Shell Jacket Reuse." *Journal of Industrial Ecology* 28, no. 3: 542–555.	Sweden
2	André, H., and Björklund, A. 2023. "A Framework to Open the Black Box of the Use Phase in Circular Economy Life Cycle Assessments: The Case of Shell Jacket Reuse." *Journal of Industrial Ecology* 27, no. 4: 1137–1150.	Sweden
3	Arpin, M. L., Leclerc, S. H., and Lonca, G. 2024. "The Circular Economy (CE) Rebound as a Paradox of Knowledge: Forecasting the Future of the CE–IoT Nexus Through the Global E-Waste Crisis." *Sustainability* 16, no. 15: 6364.	Canada and France
4	Bączyk, M., Tunn, V., Worrell, E., and Corona, B. 2024. "Consumer Behavior in Circular Business Models: Unveiling Conservation and Rebound Effects." *Sustainable Production and Consumption* 52, 283–298.	Netherlands (Utrecht University)
5	Bartoszczuk, P. 2023. "Circular Economy and Its Restriction." *Economics and Environment*.	Poland
6	Bertassini, A. C., Calache, L. D. D. R., Carpinetti, L. C. R., Ometto, A. R., and Gerolamo, M. C. 2022. "CE-Oriented Culture Readiness: An Assessment Approach Based on Maturity Models and Fuzzy Set Theories." *Sustainable Production and Consumption* 31: 615–629.	Brazil
7	Beyeler, L., and Jaeger-Erben, M. 2022. "How to Make More of Less: Characteristics of Sufficiency in Business Practices." *Frontiers in Sustainability* 3: 949710.	Germany
8	Birat, J. P. 2015. "Life Cycle Assessment, Resource Efficiency and Recycling." *Metallurgical Research & Technology* 112, no. 2: 206.	France
9	Bocken, N. M., Niessen, L., and Short, S. W. 2022. "The Sufficiency-Based Circular Economy—An Analysis of 150 Companies." *Frontiers in sustainability* 3: 899289.	Netherlands
10	Bubinek, R., Knaack, U., and Cimpan, C. 2025. "Reuse of Consumer Products: Climate Account and Rebound Effects Potential." *Sustainable Production and Consumption* 54: 190–201.	Denmark
11	Castro, C. G., Trevisan, A. H., Pigosso, D. C., and Mascarenhas, J. 2022. "The Rebound Effect of Circular Economy: Definitions, Mechanisms and a Research Agenda." *Journal of Cleaner Production* 345: 131136.	Brazil
12	Chatzistamoulou, N., and Koundouri, P. 2024. "Is Green Transition in Europe Fostered by Energy and Environmental Efficiency Feedback Loops? The Role of Eco-Innovation, Renewable Energy and Green Taxation." *Environmental and Resource Economics* 87, no. 6: 1445–1472.	European Union
13	Chatzistamoulou, N., and Tyllianakis, E. 2022. "Green Growth & Sustainability Transition Through Information. Are the Greener Better Informed? Evidence From European SMEs." *Journal of Environmental Management* 306: 114457.	European Union
14	Chen, C. W. 2021. "Clarifying Rebound Effects of the Circular Economy in the Context of Sustainable Cities." *Sustainable Cities and Society* 66: 102622.	Taiwan
15	Ciechelska, A., Matel, A., Poskrobko, T., and Sidorczuk-Pietraszko, E. 2023. "Circular Economy Rebound Effect in the Context of Second-Hand Clothing Consumption." *Economics and Environment* 87, no. 4: 635–620.	Poland
16	Csutora, M., Harangozo, G., and Szigeti, C. 2022. "Sustainable Business Models—Crisis and Rebound Based on Hungarian Research Experience." *Resources* 11, no. 12: 107.	Hungary
17	Das, A., Konietzko, J., Bocken, N., and Dijk, M. 2023. "The Circular Rebound Tool: A Tool to Move Companies Towards More Sustainable Circular Business Models." *Resources, Conservation & Recycling Advances* 20: 200185.	Netherlands

(Continues)

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

No.	References	Country
18	Fan, F., Zhang, K., Dai, S., and Wang, X. 2023. "Decoupling Analysis and Rebound Effect Between China's Urban Innovation Capability and Resource Consumption." *Technology Analysis & Strategic Management* 35, no. 4: 478–492.	China
19	Ferrante, M., Vitti, M., Facchini, F., and Sassanelli, C. 2024. "Mapping the Relations Between the Circular Economy Rebound Effects Dimensions: A Systematic Literature Review." *Journal of Cleaner Production* 456: 142399.	Italy and France
20	Fertner, C., and Große, J. 2016. "Compact and Resource Efficient Cities? Synergies and Trade-Offs in European Cities." *European Spatial Research and Policy* 23, no. 1: 65–79.	Europe
21	Figge, F., and Thorpe, A. S. 2019. "The Symbiotic Rebound Effect in the Circular Economy." *Ecological Economics* 163: 61–69.	France and Australia
22	Flachenecker, F. 2018. "The Causal Impact of Material Productivity on Macroeconomic Competitiveness in the European Union." *Environmental Economics and Policy Studies* 20, no. 1: 17–46.	European Union
23	Flores, J., Cavique, M., and Seixas, J. 2022. "Energy Sustainability—Rebounds Revisited Using Axiomatic Design." *Sustainability* 14, no. 11: 6737.	Portugal
24	Font Vivanco, D., Freire-González, J., Galvin, R., et al. 2022. "Rebound Effect and Sustainability Science: A Review." *Journal of Industrial Ecology* 26, no. 4: 1543–1563.	Europe, USA, and developing countries
25	Font Vivanco, D., Sala, S., and McDowall, W. 2018. "Roadmap to Rebound: How to Address Rebound Effects From Resource Efficiency Policy." *Sustainability* 10, no. 6: 2009.	United Kingdom and European Union
26	Freeman, R. 2018. "A Theory on the Future of the Rebound Effect in a Resource-Constrained World." *Frontiers in Energy Research* 6: 81.	United Kingdom
27	Genovese, A., Lowe, B. H., Bimpizas-Pinis, M., and Ram, V. G. 2024. "Planning for a Future Free From Rebound Effects." *Futures* 164: 103479.	United Kingdom
28	Gheewala, S. H. 2024. "Material Recycling in a Circular Economy—A Systems View." *Wiley Interdisciplinary Reviews: Energy and Environment* 13, no. 1: e503.	Thailand
29	Godinot, O., Jouan, J., Nesme, T., and Carof, M. 2024. "Evidence of a Rebound Effect in Agriculture: Crop-Livestock Reconnection Beyond the Farm Gate Does Not Always Lead to More Sustainable Nitrogen Management." *Agricultural Systems* 221: 104137.	Spain
30	Gossart, C. 2014. "Rebound Effects and ICT: A Review of the Literature." *ICT Innovations for Sustainability*, 435–448.	France
31	Grădinaru, G. I., and Maricuț, A. C. 2022. "From the Rebound Effect to the Perspective of Circular Economy: A Structure Changes Analysis Among EU Countries." *Economic Computation & Economic Cybernetics Studies & Research* 56, no. 1.	Romania
32	Henry, M., Schraven, D., Bocken, N., Frenken, K., Hekkert, M., and Kirchherr, J. 2021. "The Battle of the Buzzwords: A Comparative Review of the Circular Economy and the Sharing Economy Concepts." *Environmental Innovation and Societal Transitions* 38: 1–21.	Netherlands and Sweden
33	Huang, C., and Lin, B. 2025. "The Impact of Digital Economy on Energy Rebound Effect in China: A Stochastic Energy Demand Frontier Approach." *Energy Policy* 196: 114418.	China
34	Jaehn, F., and Meissner, F. 2022. "The Rebound Effect in Transportation." *Omega* 108: 102563.	Germany
35	Kjaer, L. L., Pigosso, D. C., McAloone, T. C., and Birkved, M. 2018. "Guidelines for Evaluating the Environmental Performance of Product/Service-Systems Through Life Cycle Assessment." *Journal of Cleaner Production* 190: 666–678.	Denmark
36	Koide, R., Murakami, S., and Nansai, K. 2022. "Prioritising Low-Risk and High-Potential Circular Economy Strategies for Decarbonisation: A Meta-Analysis on Consumer-Oriented Product-Service Systems." *Renewable and Sustainable Energy Reviews* 155: 111858.	Japan
37	Konash, A., and Nasr, N. 2022. "The Circular Economy and Resource Use Reduction: A Case Study of Long-Term Resource Efficiency Measures in a Medium Manufacturing Company." *Cleaner Production Letters* 3: 100025.	United States
38	Lahcen, B., Eyckmans, J., and Brusselaers, J. 2024. "Linking Methodologies to Assess Climate Impacts and Circular Economy Strategies Along Supply Chains." *Sustainable Production and Consumption* 47: 544–556.	Belgium

(Continues)

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE A1 | (Continued)

No.	References	Country
39	Laurenti, R., Singh, J., Cotrim, J. M., Toni, M., and Sinha, R. 2019. "Characterizing the Sharing Economy State of the Research: A Systematic Map." *Sustainability* 11, no. 20: 5729.	Sweden, Switzerland, Portugal, and Italy
40	Levänen, J., Uusitalo, V., Härri, A., Kareinen, E., and Linnanen, L. 2021. "Innovative Recycling or Extended Use? Comparing the Global Warming Potential of Different Ownership and End-of-Life Scenarios for Textiles." *Environmental Research Letters* 16, no. 5: 054069.	Finland
41	Liu, X., Wang, S., Li, Y., Ali, S., Khan, Y. A., Salem, S., and Huang, X. 2022. "RETRACTED: Recycling in Textile Sector: A New Circular Economy Approach Towards Ecology and Environmental Sustainability." *Frontiers in Environmental Science* 10: 929710.	Several countries with a focus on Europe and China
42	Lowe, B. H., Bimpizas-Pinis, M., Zerbino, P., and Genovese, A. 2024. "Methods to Estimate the Circular Economy Rebound Effect: A Review." *Journal of Cleaner Production* 443: 141063.	Several countries, including the United Kingdom, the United States, China, and Switzerland
43	Makov, T., and Font Vivanco, D. 2018. "Does the Circular Economy Grow the Pie? The Case of Rebound Effects From Smartphone Reuse." *Frontiers in Energy Research* 6: 39.	United States
44	Mallari, C. B., Li, R., and San Juan, J. L. 2024. "A System Dynamics Model for Analyzing the Circular Economy Rebound." *Chemical Engineering Transactions* 114: 337–342.	Philippines
45	Mata, É., Kihila, J. M., Wanemark, J., et al. 2022. "Non-Technological and Behavioral Options for Decarbonizing Buildings—A Review of Global Topics, Trends, Gaps, and Potentials." *Sustainable Production and Consumption* 29: 529–545.	Sweden with collaborations from Tanzania and the United States
46	Metic, J., and Pigosso, D. C. 2022. "Research Avenues for Uncovering the Rebound Effects of the Circular Economy: A Systematic Literature Review." *Journal of Cleaner Production* 368: 133133.	Denmark
47	Metic, J., Guzzo, D., Kopainsky, B., McAloone, T. C., and Pigosso, D. C. 2024. "A Simulation-Based Approach for Investigating the Dynamics of Rebound Effects in the Circular Economy: A Case of Use-Oriented Product/Service System." *Journal of Environmental Management* 365: 121627.	Denmark and Norway
48	Morimoto, S., Kuroki, H., Narita, H., and Ishigaki, A. 2021. "Scenario Assessment of Neodymium Recycling in Japan Based on Substance Flow Analysis and Future Demand Forecast." *Journal of Material Cycles and Waste Management* 23, no. 6: 2120–2132.	Japan
49	Niehoff, S., and Beier, G. 2018. "Industrie 4.0 and a Sustainable Development: A Short Study on the Perception and Expectations of Experts in Germany." *International Journal of Innovation and Sustainable Development* 12, no. 3: 360–374.	Germany
50	Niero, M., Jensen, C. L., Fratini, C. F., Dorland, J., Jørgensen, M. S., and Georg, S. 2021. "Is Life Cycle Assessment Enough to Address Unintended Side Effects From Circular Economy initiatives?" *Journal of Industrial Ecology* 25, no. 5: 1111–1120.	Denmark
51	Onat, N. C., Mandouri, J., Kucukvar, M., et al. 2023. "Rebound Effects Undermine Carbon Footprint Reduction Potential of Autonomous Electric Vehicles." *Nature Communications* 14, no. 1: 6258.	Qatar
52	Ottelin, J., Cetinay, H., and Behrens, P. 2020. "Rebound Effects May Jeopardize the Resource Savings of Circular Consumption: Evidence From Household Material Footprints." *Environmental Research Letters* 15, no. 10: 104044.	Europe
53	Piscicelli, L. 2023. "The Sustainability Impact of a Digital Circular Economy." *Current Opinion in Environmental Sustainability* 61: 101251.	Netherlands
54	Pohl, J., Hilty, L. M., and Finkbeiner, M. 2019. "How LCA Contributes to the Environmental Assessment of Higher Order Effects of ICT Application: A Review of Different Approaches." *Journal of Cleaner Production* 219: 698–712.	Germany and Switzerland
55	Rabiu, M. K., and Jaeger-Erben, M. 2022. "Appropriation and Routinisation of Circular Consumer Practices: A Review of Current Knowledge in the Circular Economy Literature." *Cleaner and Responsible Consumption* 7: 100081.	Germany and Denmark
56	Roa, D., and Rosendahl, K. E. 2023. "Policies for Material Circularity: The Case of Lithium." *Circular Economy and Sustainability* 3, no. 1: 373–405.	Norway

(Continues)

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE A1 | (Continued)

No.	References	Country
57	Safarzynska, K., Di Domenico, L., and Raberto, M. 2023. "The Circular Economy Mitigates the Material Rebound due to Investments in Renewable Energy." *Journal of Cleaner Production* 402: 136753.	Poland and Italy
58	Saha, K., Dey, P. K., and Kumar, V. 2024. "A Comprehensive Review of Circular Economy Research in the Textile and Clothing Industry." *Journal of Cleaner Production* 444: 141252.	United Kingdom
59	Salvador, R., Barros, M. V., Donner, M., Brito, P., Halog, A., and De Francisco, A. C. 2022. "How to Advance Regional Circular Bioeconomy Systems? Identifying Barriers, Challenges, Drivers, and Opportunities." *Sustainable Production and Consumption* 32: 248–269.	Multinational (18 countries, including Brazil, Portugal, and the United States)
60	Salvador, R., Puglieri, F. N., Halog, A., de Andrade, F. G., Piekarski, C. M., and De Francisco, A. C. 2021. "Key Aspects for Designing Business Models for a Circular Bioeconomy." *Journal of Cleaner Production* 278: 124341.	Brazil with international collaborators
61	Samani, P. 2025. "Replaying the Circular Economy Orchestra: The Echoes of R and the Symphony of I." *Journal of Integrative Environmental Sciences* 22, no. 1: 2440156.	Netherlands
62	Santarius, T., Walnum, H. J., and Aall, C. 2018. "From Unidisciplinary to Multidisciplinary Rebound Research: Lessons Learned for Comprehensive Climate and Energy Policies." *Frontiers in Energy Research* 6: 104.	Germany and Norway
63	Schultz, F. C., Valentinov, V., Reinhardt, R. J., and Pies, I. 2024. "The Circular Economy Rebound Effect: Reconceptualizing Rebound Approaches and Mitigation Opportunities From an Ordonomic Perspective." *Journal of Industrial Ecology* 28, no. 3: 374–385.	Germany
64	Shah, A. A., and Zha, D. 2025. "Economy-Wide Estimates of the Energy Rebound Effect in BRICS: The Role of Environmental Regulations and Economic Policy Uncertainty." *Energy* 320: 135276.	BRICS (Brazil, Russia, India, China, South Africa)
65	Sheng, X., Zhang, X., and Zhou, X. 2023. "Show Me the Impact: Communicating "Behavioral Impact Message" to Promote Pro-Environmental Consumer Behavior." *Sustainable Production and Consumption* 35: 709–723.	China/Hong Kong
66	Shinde, R., Froemelt, A., Kim, A., and Hellweg, S. 2022. "A Novel Machine-Learning Approach for Evaluating Rebounds-Associated Environmental Footprint of Households and Application to Cooperative Housing." *Journal of Environmental Management* 304: 114205.	Switzerland
67	Siderius, T., and Poldner, K. 2021. "Reconsidering the Circular Economy Rebound Effect: Propositions From a Case Study of the Dutch Circular Textile Valley." *Journal of Cleaner Production* 293: 125996.	Netherlands
68	Sourabh, S., Menon, B. G., and Mahanty, B. 2024. "Econometric Analysis of Circular Economy Co-Flow Process in Metal Industry." *Quality & Quantity* 58, no. 2: 1583–1602.	India
69	Spiller, M., Vingerhoets, R., Vlaeminck, S. E., Wichern, F., and Papangelou, A. 2024. "Beyond Circularity! Integration of Circularity, Efficiency, and Sufficiency for Nutrient Management in Agri-Food Systems." *Nutrient Cycling in Agroecosystems* 129, no. 3: 287–297.	Belgium
70	Tsuji, K., Kurisu, K., Nakatani, J., and Moriguchi, Y. 2020. "Evaluation of Environmental Impact of Car Sharing in Consideration of Uncertainty of Influential Variables." *International Journal of Automation Technology* 14, no. 6: 975–983.	Japan
71	Vegter, D., van Hillegersberg, J., and Olthaar, M. 2023. "Performance Measurement System for Circular Supply Chain Management." *Sustainable Production and Consumption* 36: 171–183.	Netherlands
72	Wallenborn, G. 2018. "Rebounds Are Structural Effects of Infrastructures and Markets." *Frontiers in Energy Research* 6: 99.	Belgium
73	Warmington-Lundström, J., and Laurenti, R. 2020. "Reviewing Circular Economy Rebound Effects: The Case of Online Peer-to-Peer Boat Sharing." *Resources, Conservation & Recycling*: X 5: 100028.	Finland
74	Wiprächtiger, M., Rapp, M., Hellweg, S., Shinde, R., and Haupt, M. 2022. "Turning Trash Into Treasure: An Approach to the Environmental Assessment of Waste Prevention and Its Application to Clothing and Furniture in Switzerland." *Journal of Industrial Ecology* 26, no. 4: 1389–1405.	Switzerland
75	Zerbino, P. 2022. "How to Manage the Circular Economy Rebound Effect: A Proposal for Contingency-Based Guidelines." *Journal of Cleaner Production* 378: 134584.	Italy

15353966, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70181, Wiley Online Library on [24/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License